Certifiable Trajectory Optimization for Locomotion via Semidefinite Relaxation

Scientific thesis for the procurance of the degree M.Sc.
(Intermediate report)
from the TUM School of Computation, Information and Technology
at the Technical University of Munich.

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In your final hardback copy, replace this page with the signed exercise sheet.

!! If you delete this page, make sure that the page structure is maintained !!
(when printed, (even) page numbers on *left pages* must be on the left side, (odd) page numbers on *right pages* on the right side

Before modifying this document, READ THE INSTRUCTIONS AND GUIDELINES!
(remove *tutorial* flag in the first line of this file to remove template tutorial parts)

Abstract

A short (1–3 paragraphs) summary of the work. Should state the problem, major assumptions, basic idea of solution, results. Avoid non–standard terms and acronyms. The abstract must be able to be read completely on its own, detached from any other work (e.g., in collections of paper abstracts). Do not use references in an abstract.

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Chapter 1

Working with this Template

Please carefully read the following instructions about student reports and specifically this template. The contents of Chap. 1 summarize the basics about working with this template. The subsequent chapters focus on specific parts of your report, e.g., the introduction (Chap. 2).

1.1 General Remarks

Please have a look at our thesis-guidelines before submitting your *final* thesis.

1.1.1 Revisions of your thesis

Hand in your thesis at minimum **one week** before the deadline for correction. You will receive feedback for the final version and very likely have to do minor or major revisions of your writing. Plan your writing schedule to allow for these adjustments, which can have quite some impact on your grade!

1.1.2 Structure of your thesis

Make sure your thesis is well structured, that each major section does what it is supposed to do, and that the overall story is clear. This template offers a common basic structure (but other structures are possible). In particular, you do not need to have exactly as many major sections or chapters as the template implies; sometimes it makes sense to merge parts, sometimes it makes sense to move parts (e.g., the location of the literature review may vary), sometimes it makes sense to split a logical part into several individual sections. Just use some common sense.

You do not have to write down everything you do. Focus on the main results of your work and structure your report accordingly. The goal is a structured report that provides everything needed to understand and replicate your work (data, formulas,

parameter values, ...). Not everything that you do/read/plot during your work has potential to be put in the report. A longer report is not necessarily better!

A sample structure is briefly provided in the following. Note again that your report does not necessarily have to follow this sample structure perfectly.

(Sample) Structure in a nutshell:

- Chapter 1: Introduction
  - General motivation - understandable even to your grandma.
  - What has been done before? (related work: How do other people solve your problem? What are drawbacks of their approaches?)
  - How do you approach the issue? What are the challenges? What methods do you use? What is new about your approach?
  - Describe the structure of your report:
  - This report is structured as follows: Chap. 2 introduces .... In Chap. 3 .... is discussed in Chap. 4. ... and so on ...
  - Optional: put notation here.
- Chapter 2: Relevant methods - describe the methods and structures you use
- Chapter 3: Describe your approaches
- Chapter 4: Simulation results
  - Describe the setup/model/implementation.
  - Give all parameters necessary to replicate your results.
  - Show results.
  - Discuss results thoroughly (a critical view point is important here).
- Chapter 5: Experimental results (structure similar to simulation results)
- Chapter 6: Conclusion
  - Summarize the goal of your work.
  - Summarize your approach and your results.
  - Draw conclusions.
  - Give an outlook on possible future work.

1.2 Structuring: Sections

Use chapters, sections, and subsections to structure your document.

1.2.1 Subsection

This is a subsection.

Subsubsection

Subsubsections can be used to further structure information within a subsection. They do not show up in your list of contents. Please do not use more than these two "sub"-prefixes to sections (no: subsubsubsubsection). If you feel that you need even finer grained structuring, rethink your overall structure.

Paragraphs are sometimes more useful than subsubsections to structure text without breaking the flow.

1.3 Style and Expressions

Before handing in your thesis, even for an intermediate review, please perform a spellcheck and correct grammar mistakes. Proof-read your work thoroughly BEFORE you send it to your supervisor. Text/latex editors usually have a spell check option included - use it! Be aware that latex editors often do not have the most sophisticated spellchecks.

The report is not meant to be a narrative text. Please stick to neutral and technical style and avoid subjective or biased expressions or adjectives/adverbs such as *obviously, always, very, especially well, actually, so-called etc.* Scientific writing is about precision and you should underpin your statements factually, not soften them with unnecessary qualifiers. Look at the papers you read and use them as a reference on how other people write scientifically.

If you know that you often make the same errors without meaning to (e.g., use past tense, forget protected spaces, use present progressive, use a word in the wrong way, ...) in many cases the search (or replace) function helps in finding them all.

1.3.1 Writing

Choose whether you want to write American English or British English and stick with it (more common is American English). If your spellcheck allows to set dictionaries, pick the appropriate language pack (British/US) to get corrections hints that help you to consistently write in either of the two languages. Make sure you are aware of the basic rules of English grammar (sentence structure, adverbs with "-ly" at the end, use of punctuation, etc.).

Keep it simple:

- Simple and short sentences (subject - verb - object)
- Simple words you know even if it sounds repetitive (Your teacher probably told you in school, that you should find synonyms whenever possible. You can do this when you write in your native language and you have a feeling for the words. Otherwise it is okay to sound repetitive, especially for nouns. And in any case: Do NOT just use new words without thoroughly researching all their meanings, really, don't!)
- If you really cannot avoid using new words, double-check whether the word really means what you want to express (translate the word back into your mother tongue and check its other meanings)

Do not use "I", use "we" instead, but do not overuse as this may sound too casual for your report. Instead use a passive form or activated objects (e.g. "The figure shows ...", "The robot moves ..."). Use present tense without exceptions (especially no past tense - NEVER - not even when you write about related work or in the conclusion). Instead use simple present ("We write this in our report.") - never present progressive ("-ing" forms: "We are not writing this in our report."). Also keep in mind that punctuation is not optional. Line breaks, on the other hand, should only be used whenever a new train of thought starts. Do not overuse them. Articles (the/a/an) are also NOT optional. If you are not sure which preposition (in, on, to, ...) to use with a word, remember: Some words are used with various prepositions, but then they often have different meanings! Do not overuse "can" or "cannot". There are other formulations that often fit better.

1.4 References

Use references to your equations/theorems/figures/etc. to make it easier for the reader to follow, especially when you need it again later in a different context. References to chapters/sections/subsections are labeled with `\label{...}` and referred to with `\ref{...}`. Use:

- Capital "Chapter`~\ref{...}`" for chapters or "Section`~\ref{...}`" for sections and subsections with protected space "~" at the beginning of sentences:
  - Chapter 1/Section 1.1 describes . . . .
- Capital "Chap.`~\ref{...}`" for chapters or "Sec.`~\ref{...}`" for sections and subsections with protected space "~" within sentences:
  - . . . is described in Chap. 1/Sec. 1.1.

- Small “chapter” or “section” (NEVER subsection, . . . - these are also sections) if it is not followed by a reference:
  - In the following chapter/section, the method . . . .

1.4.1 Equations

Refer to equations whenever it helps in understanding your text. This makes it easier for the reader to follow your train of thought:

- NEVER refer to an equation before you introduce it.
- Equation numbers should be put in brackets (1.1) - use`~\eqref{}`, which does it automatically for you. Use the protected space “~” to avoid lines starting with equation numbers.
- Do not start a sentence with an equation number.
- Do not write “equation” before putting the reference:
  - Bad: Given Equation (1.1), it is possible to show . . . .
  - Good: Given (1.1), it is possible to show . . . .

1.4.2 Figures

References to figures should be included in the following way:

- Capital “Figure`~\ref{fig}`” with protected space “~” at the beginning of sentences:
  - Figure 1.1 shows the motion of the point mass.
- Capital “Fig.`~\ref{fig}`” with protected space “~” within sentences:
  - The motion of the point mass is shown in Fig. 1.1.
- Small “figure” if it is not followed by a reference:
  - The figure also shows . . . .

1.4.3 Tables

Tables make it easy to understand the content. Less lines are sometimes better and look nicer, take a look in literature. References to tables should be included in the following way:

- Capital “Table`~\ref{tab}`” with protected space “~” at the beginning of sentences:

  - Table 1.1 lists . . . .

- Capital "Tab.~\ref{tab}" with protected space "~" within sentences:
  - . . . are given in Tab. 1.1.

- Small "table" if it is not followed by a reference:
  - The table also lists . . . .

1.4.4 Definitions / Assumptions / Theorems / Lemmas

Basically, the same applies as for figures, tables, chapters, and sections:

- Use protected spaces "text~\ref{...}".
- Capital and full word (Definition/Assumption/Theorem/Lemma/Proposition~\ref{}) at the beginning of sentences.
- Capital and short form (Def./Ass./Th./Lem./Prop.~\ref{}) in the middle or at the end of sentences.
- Small and full word (definition/assumption/theorem/lemma/proposition) if used without reference.

1.5 Equations

Include equations in your text in a meaningful way. Example:

The equation of motion of a point mass is given by

$$m\ddot{x} = f \ , \tag{1.1}$$

where $m \in \mathbb{R}$ is the mass, $f \in \mathbb{R}$ the externally applied force, and $\ddot{x} \in \mathbb{R}$ the acceleration in x-direction.

You can use (1.1) to reference an equation in the text. Equations without reference number:

$$y = x^2$$

Please type all numbers in mathematics mode. Mathematical equations that are not part of the text should be centered. If you have multiple lines with equations, align equations with the arithmetic operator (=,<, ...):

$$a = 2 \tag{1.2a}$$

$$b = 3 + 4 \tag{1.2b}$$

Type physical units as normal text, e.g., s, not s! (the package *siunitx* helps) Formulas must not jut out over the text margin, so take care to stay within the margins and eventually break the equation into several lines. Explain the variables you use. For each newly introduced variable, at least a short description of the variable should follow the equation in the style of *...with x being the position of the cart and u the control input.* Use variable names whenever possible to make it easier for the reader to make the connection:

- Bad: A high mass slows down the acceleration.
- Good: A high mass m slows down the acceleration.

A parameter or variable must not be used at the beginning of a sentence. Mathematical units must be consistently labelled with only one variable. Do not use a variable for more than one mathematical unit.

If you use in-line equations, either to introduce parameters or for actual equations, put a "~" in between the in-line equation and the preceding word. This is a protected space - it avoids that the equation is moved to the next line, which would be considered bad style.

If you use descriptive text in equations, set it as `\text{}`. Examples:

- x_{measured}: "measured" is descriptive here and not a variable/parameter
- $q_{\text{des},i}$: "des" is descriptive here but "i" is a counter and therefore not in text

If you want to generate more sophisticated equations, use the *IEEEtrantools* package. This package is especially useful if custom spacing is required for multiple equation lines.

1.5.1 Notation Consistency

Consistency is key: Choose a style for your notation and stick with it. We suggest giving a notation list somewhere in your report (ideally in the beginning). Most often used:

- vectors: bold, small letters `\mathbf{}` ($\mathbf{a}$) or `\boldsymbol{}` ($\boldsymbol{a}$)
- matrices: bold, capital letters `\mathbf{}` ($\mathbf{A}$) or `\boldsymbol{}` ($\boldsymbol{A}$)
- scalars: small (sometimes also capital), non-bold letters (a)
- general sets: `\mathcal{}`
- sets of numbers (real, whole, ...): `\mathbb{}` ($\mathbb{A}$)

1.5.2 Symbol Glossary

Instead of writing mathematical symbols individually in each equation, you can also use a glossary entry. For example, instead of explicitely writing `\mathbf{u}_k` to denote a control input $\mathbf{u}_k$ at time step k, you can define a glossary entry, e.g., "uk", and just use `\gls{uk}`, which then also yields $\mathbf{u}_k$. Glossary entries for symbols are defined in the file ./include/gloss.tex.

This procedure is useful for two reasons: 1) You can shorten long symbol expressions significantly. 2) If you later want to change a symbol, you only need to change the glossary entry once - you don't need to change every single symbol in your document.

1.5.3 Style Notes

There are a few important style notes to consider in equations.

- Indices: Use italic font for math-related indices (parameters, variables, ...) and non-italic font for other descriptions, e.g., x_t (time t) and x_{ref} (ref: reference).
- Mathematics operations are non-italic, e.g., $\sin(x)$. Often, functions are already defined in latex (e.g., use `\sin(x)`). If not, use `\mathrm{}`, e.g., $\sin(x)$.
- Make sure parentheses are large enough by using `\left( . \right)`, e.g.,

$$\left(\frac{1}{2}\right) \text{ instead of } (\frac{1}{2}).$$

- Use `^\top` for a transpose, e.g., $\boldsymbol{x}^\top$.

1.6 Figures

For pictures, you can use .pdf, .jpg, .png, etc when working with **pdflatex**; with **latex** you need .eps:

Figure 1.1: Long Description: The subtitles of tables and illustrations should be self-explanatory.

Figures should be vector graphics (pdf or eps) and self-made (to avoid blurry pictures). We recommend pgfplots/tikzpiktures or inkscape and Matlab (export figure with tick set at Rendering - Custom renderer: painters) in combination with `\psfrag{ }[ ][ ]{ }` to replace the text in LaTeX.

Every illustration must be described and referenced in the text, cf. Fig. 1.1. There is no such thing as a decorative figure! If the figure is not mentioned in the text and therefore lacks an explanatory function, leave it. Lines in plots need to

be clearly distinguishable, also in black-and-white print. Therefore, choose your colors and line styles wisely and consistently! The subtitles and axis labels must be clearly legible (i.e., big enough) and include a scaling. Ideally, the figure is self-explanatory with labels, arrows, text, etc. Text in figures should be without serif: `{\sf your_text_goes_here}`, about the same size as the "normal" text and labels should include all important information such as the name of the depicted variable and the corresponding unit. Captions should clearly describe the content. If the caption is rather long, use "`\caption[short_caption]{long caption}`". The long caption appears below the figure, the short caption appears in the list of figures.

Figure axes need to be labeled correctly. Do not just mention the variable name but include a description. If units are required, use parenthesis. An example is given in Fig. 1.2.

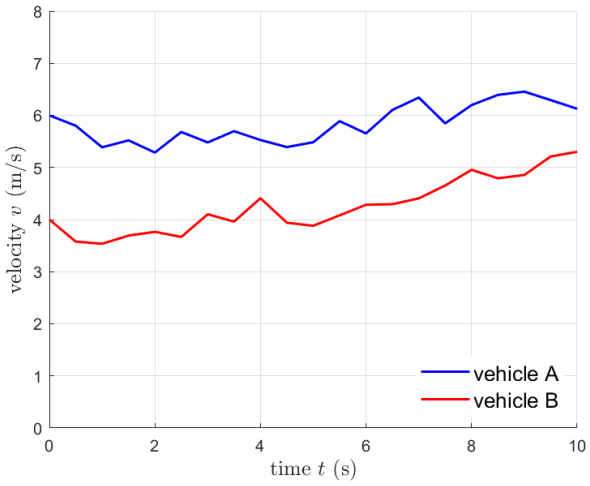


Figure 1.2: Example velocity plot with units

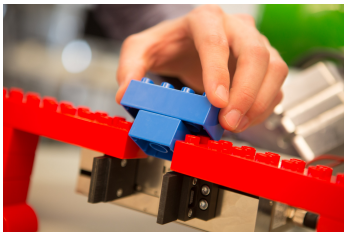
If the illustration is copied from another person's work, you need to mention the source with `\cite{}`.

By using *subfigure*, several pictures can be combined in one main picture as in Fig. 1.3:

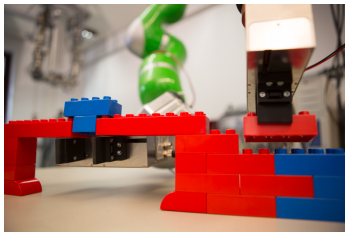
This figures can be referenced by either referencing the overall label 1.3 or a specific subfigure label 1.3(a) - if the labels have been set properly of course! The subfigures are not listed in the list of figures!

1.7 Definitions / Assumptions / Theorems / Lemmas

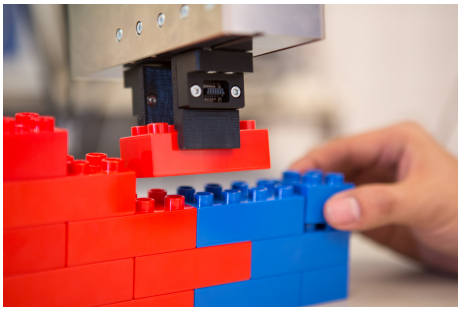
Sometimes a definition is helpful, especially for later reference. Motivate and introduce each one and do not just list definitions one after another. Explain why



(a) Subfigure 1 caption



(b) Subfigure 2 caption



(c) Subfigure 3 caption

A Picture needs to be put here later. One can use acronyms, e.g. human-robot collaboration (HRC)

(d) Space-holder for missing picture

Figure 1.3: A figure with two subfigures

it is important and what it is used for. Assumptions are useful to emphasize what preliminaries have to be fulfilled for your approach to work, especially if you plan to introduce theorems, lemmas, etc.

Points to remember:

- Do not list assumptions one after another.
- Motivate and introduce each one.
- Explain how restrictive it is and illustrate its effects on the application.

Sometimes you need to cite a theorem/lemma/proposition.
Points to remember:

- Same as above: do not just list, motivate, introduce, explain what this enables.
- Make yourself familiar with the way theorems/... are usually formulated. Look at the literature.
- Make sure all relevant assumptions and preliminaries are stated in the theorem/...
- Make the proof (if it is your own theorem/...) conclusive and understandable.

1.8 Citations

Citations are extremely important and **must** be included whenever you use other people's work. Direct citations with directly copy-pasting the content and marking it with "..." are unusual in scientific reports. Always try to reformulate the source into your own words to generate an indirect citation.

Whether to use numeric or alphabetic references is not all that important (unless prescribed by a conference or journal), but alphabetic tends to be more readable. Independent of citation style, the following rules should be followed:

- Use the LaTeX cite package. It does not give you additional commands, but it fixes a few quirks in LaTeX. Among others, it automatically sorts multiple citations, and it correctly spaces the angular brackets (if you use the `\cite{}` command without leading white space).

- Citing several papers at one point should be done with a single `\cite{}` command. For example, use `gives good results\cite{Bloe_99, Jay_87}`, resulting in gives good results [?, ?].

  Do not use `gives good results\cite{Bloe_99}\cite{Jay_87}`, which produces the ugly gives good results [?][?]. Also, note that there is no space between the `\cite{}` command and the preceding word, LaTeX (with the cite package) does the spacing correctly.

- Avoid citations of the kind: [?] thinks that $x > y$ is valid, but [?] argues that this is invalid in case of $z \geq 5$. This works a bit better if using alphanumeric citation labels. Better, though, use the authors' names: Bloe and Joe [1] think that $x > y$ is valid, but O'Neill et al. [?] argues that this is invalid in case of $z \geq 5$.

- Avoid citations at the beginning of a sentence:
  - Bad:
    - [?] introduces different methods for motion control. (citation at the beginning)
    - Different methods for motion control are introduced in literature. [?] (citation after the punctuation)
    - Different methods for motion control are introduced in[?]. (preceding space missing or wrong cite package)
  - Good:
    - Different methods for motion control are introduced in [?].
    - Khalil et al. introduce different methods for motion control [?].
    - Motion control as in [?] is generally used to . . . .
    - Motion control [?] is generally used to . . . .

    * Motion control is generally used to ... [**?**].

- BibTeX is a great tool, but you need to know how to use it. A regular trap is to forget that TEX knows more about typesetting than you do. So, for example, it changes the case of words in the title. If your title contains acronyms and proper names (most do), they tend to get down-cased. Any such words which should not have their case changed should be put into braces, e.g., `{The {Mungi} {OS} and its Use in Merry-Go-Round Seat Allocation}`.

- In citations do not abuse the category technical report. People tend to cite just about anything that has not been published in a journal or conferences as a TR. This is outright wrong. The concept of a TR is actually fairly well defined: A TR is published in some sort. This is generally as part of a formal TR series of some institution, in hardcopy or on the web or both. (They are not always called "technical report", other common names are "research report", "technical memorandum", <institution> "report" etc.) The publication (i.e., availability outside) is essential, otherwise it's at best an internal report. A TR has a number (absolutely!), an institution (publisher), a date (month and year at least) and a publisher's address (besides all the other stuff bibentries have). If your document doesn't have these features, it's not a TR. It is probably better categorized as a working paper. Even then it has a date and an institution address.

- Citing web pages is often unavoidable (but also often a sign of laziness). When citing web pages be aware that they may only be short-lived. Consider whether the reference will be of any use to the reader at all if the link is broken. Or whether your whole document only has a use-by date a few months past writing.

Any cited document, whatever it may be, has a few **mandatory** features:

- Date. Absolutely.

- Author/organization/creator/person responsible for contents.

- Whatever information the reader needs to find that document. In most BibTeX entry types these are clearly identified as mandatory fields. Mandatory means that they aren't optional. Do not pretend they are. For a working paper these might be the contact details of the author.

For bibliography, edit `mybib.bib` and list all references in a special style, e.g., for a book:

```
@book{literaturstelle1,
 author    = {S. Sastry},
 title     = {Nonlinear Systems - Analysis, Stability, and Control},
```

```
 publisher = {Springer},
 year      = 1999
}
```

Cite references in the text by `\cite{citationreference}`.

In order to have your references shown in your PDF, compile by:

```
latex myfile
bibtex myfile
latex myfile
latex  myfile
dvips myfile
ps2pdf -sPAPERSIZE=a4 myfile.ps
```

Alternatively you can use your preferred LaTeX-editor. The according BibTeX compiler has to be configured properly and then run before running PdfLaTeX. We recommend you to use our preconfigured editors.

Make sure your bibliography is consistent (e.g., all first names abbreviated).

1.9 Compiling

Linux: do the following in a Shell:

```
latex myfile
dvips myfile
ps2pdf -sPAPERSIZE=a4 myfile.ps
```

Make sure to use the A4 option, when you print the pdf!

Alternatively you can use either Kile, Texmaker, etc. on Linux or TeXnic Centre, etc. on Windows. Please be aware you have to set the required LaTeXpaths in your editors. The most common LaTeX-editors in Linux are preconfigured at LSR and should work *out-of-the-box*.

Chapter 2

Introduction

Your first chapter in the document. Introduce the problem (gently!). Try to give the reader an appreciation of the difficulty, and an idea of how you will go about it. It is like the overture of an opera: it plays on all the relevant themes.

Make sure you clearly state the vision/aims of your work, which problem you are trying to solve, and why it is important. While the introduction is the part that is read first (ignoring title and abstract), it is usually best written last (when you actually know what you have really achieved). Remember, it is the first thing that is read and will have a major influence on the how the reader approaches your work. If you bore them now, you have most likely lost them already. If you make outrageous claims, pretend to solve the world's problems, etc, you are likely fighting an uphill battle later on. Also, make sure you pick up any threads spun in the introduction later on, to ensure that the reader thinks they get what they have been promised. Do not create an expectation that you will deliver more than you actually do. Remember, the reader may be your marker (of a thesis) or referee (of a paper), and you do not want to annoy them.

2.1 Problem Statement

You can either state the problem you are trying to solve in the general introduction, providing the transition from the overall picture to your specific approach, or state it in a separate section. Even if you do not use the separate section, writing down in a few sentences why the problem you are trying to solve is actually hard and has not been solved before can give you a better idea of how to approach the topic. This can be also merged with the related work part.

2.2 Related Work

From Kevin Elphinstone's *A Small Guide to Writing Your Thesis* [**?**]:

"The related work section (sometimes called literature review) is just that, a review of work related to the problem you are attempting to solve. It should identify and evaluate past approaches to the problem. It should also identify similar solutions to yours that have been applied to other problems not necessarily directly related to the one your solving. Reviewing the successes or limitations of your proposed solution in other contexts provides important understanding that should result in avoiding past mistakes, taking advantage of previous successes, and most importantly, potentially improving your solution or the technique in general when applied in your context and others.

In addition to the obvious purpose indicated, the related work section also can serve to:

- justify that the problem exists by example and argument
- motivate interest in your work by demonstrating relevance and importance
- identify the important issues
- provide background to your solution

Any remaining doubts over the existence, justification, motivation, or relevance of your thesis topic or problem at the end of the introduction should be gone by the end of related work section.

Note that a literature review is just that, a review. It is not a list of papers and a description of their contents! A literature review should critique, categorize, evaluate, and summarize work related to your thesis. Related work is also not a brain dump of everything you know in the field. You are not writing a textbook; only include information directly related to your topic, problem, or solution."

Note: Write literature review at an early stage of your project to build on the knowledge of others, not reinvent the wheel over and over again! There is nothing more frustrating after weeks or months of hard work to find that your great solution has been published 5 years ago and is considered old news or that there is a method known that produces superior results.

Chapter 3

Technical Approach

3.1 Design of your solution

Having explained the problem, and what others have done in similar situations, now explain your approach. Again, give a general overview of your design first, and then go into detail. The important part here is the concept of your work, not the actual implementation! Make sure that the document (particularly a thesis) is self-contained: It should be possible for a reader familiar with the general area to understand your design. Again, be forthright about the limitations of your design. Also, make sure you justify any shortcuts/limitations convincingly.

3.2 Implementation

In many (not all cases) there is a clear difference between the general approach (design) and its implementation in your particular circumstances. The design may be more general than what you can do given time and resources. Or you have developed a general design, and are now implementing a prototype on particular hardware. Give all required details. It should be possible to understand all this without referring to the source code.

This will, in general, include extracts of actual algorithms and hardware components used. Do not list pages of C code, an electronic copy of the source will accompany the submission and should be available to the marker, so there is no point in killing extra trees to put it into the report. Source code, if included at all, goes into the appendix and not the main document.

Make sure you describe your implementation in enough detail. Someone who has nothing else but your thesis report to go by should be able to repeat your work, and arrive at essentially the same implementation. Reproducibility is an important component of scientific work. Also, clearly state the limitations of your implementation, and justify them.

Chapter 4

Preliminaries On Special Embedded Matrix Lie Groups

4.1 Special Matrix Lie Groups

4.2 Special Lie Algebras

Proposition 1 (Differential of left translation on a matrix Lie group: T_xL_h)**.** *Let $G \subset GL(n, \mathbb{R})$ be a matrix Lie group. For $h \in G$, define $L_h : G \to G$ by $L_h(g) = hg$ [Hal00]. Then for each $x \in G$ the differential*

$$T_xL_h : T_xG \to T_{hx}G,$$

of the left translation map $L_h : G \to G$ is given by

$$T_xL_h(v) = hv. \tag{4.1}$$

Proof. By the definition of the tangent space of an embeded submanifold given in [Hal00], there exists a smooth curve $\gamma : (-\epsilon, \epsilon) \to G$ with $\gamma(0) = x$ and the tangent vector $\dot{\gamma}(0) = v \in T_xG$. In [**?**], the differential is defined as

$$T_xL_h(v) = \frac{d}{dt}L_h(\gamma(t))\bigg|_{t=0} = \frac{d}{dt}(h\gamma(t))\bigg|_{t=0},$$

with the left translation map $L_h(g) = hg$. Since h is constant and $\dot{\gamma}(0) = v$, it follows that

$$T_xL_h(v) = h\dot{\gamma}(0) = hv.$$

□

It can be observed that $L_h(g) = hg$ and $T_xL_h(v) = hv$ have the same matrix expression, but acting on different objects. The left translation map $L_h(g)$ is multiplying group elements $g \in G$, while the differential of the left translation map is the pushforward of tangent vectors $v \in T_xG \subset \mathbb{R}^{n\times n}$. The differential of a left translation can be applied to the special embedded matrix Lie groups $SO(n)$, $SE(n)$ and their corresponding lie algebras $\mathfrak{so}(n)$, $\mathfrak{se}(n)$.

Differential of left translation $T_R L_H$ on $SO(n)$ The left translation map in $SO(n)$ is defined as $L_H(R) = HR$ with $H, R \in G := SO(n)$. Applied to eq. 4.1 with $v \in T_R G$, it follows that

$$T_R L_H(v) = Hv.$$

The tangent vector $v \in T_R G$ can be represented by $v = T_e L_R(\omega) = R\omega$ using the left translation map of $\omega \in \mathfrak{g} := \mathfrak{so}(n)$ resulting in

$$T_R L_H(R\omega) = H(R\omega). \tag{4.2}$$

Differential of left translation $T_g L_h$ on $SE(n)$ The left translation map in $SE(n)$ is defined as $L_h(g) = hg$ with $h, g \in G := SE(n)$. Applied to eq. 4.1 with $v \in T_g G$, it follows that

$$T_g L_h(v) = hv.$$

The tangent vector $v \in T_g G$ can be represented by $v = T_e L_g(\xi) = g\xi$ using the left translation map of $\xi \in \mathfrak{g} := \mathfrak{se}(n)$ resulting in

$$T_g L_h(g\xi) = h(g\xi). \tag{4.3}$$

Explain brackets Inner product -¿ Pairs

Chapter 5

Continuous Mechanics of Rigid Bodies on Lie Groups

5.1 Lagrangian Mechanics

5.1.1 Euler-Lagrange Equations

$$\textit{Euler} \tag{5.1}$$

Proposition 2 (Directional derivative representation of $D_\xi L(g,\xi)$)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$ [Hal00]. Let $L : G \times \mathfrak{g} \to \mathbb{R}$ be smooth. For all $\delta\xi \in \mathfrak{g}$ we regard the partial derivative $D_\xi L(g,\xi) \in \mathfrak{g}^*$ as the covector characterized by*

$$\langle D_\xi L(g,\xi), \delta\xi\rangle = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(g, \xi + \varepsilon\,\delta\xi). \tag{5.2}$$

Proof. Since $\mathfrak{g}$ is a vector space, the differential of the map $L(g,\xi)$ at ξ is a linear map $\mathfrak{g} \to \mathbb{R}$, as g is fixed. Therefore the partial derivative $D_\xi L(g,\xi)$ is an element of the dual space $\mathfrak{g}^*$. Moreover, for any $\delta\xi \in \mathfrak{g}$, the curve $\gamma(\varepsilon) = \xi + \varepsilon\,\delta\xi$ satisfies $\gamma(0) = \xi$ and $\dot\gamma(0) = \delta\xi$, which realizes $\delta\xi$ as a tangent direction at ξ for any $\varepsilon \in (-c, c)$ [Lee08]. The corresponding directional derivative is

$$\langle D_\xi L(g,\xi), \delta\xi\rangle = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(g,\gamma(\varepsilon)) = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(g, \xi + \varepsilon\,\delta\xi),$$

using $\langle D_\xi L(g,\xi), \delta\xi\rangle = D_\xi L(g,\xi)(\delta\xi)$. □

Proposition 3 (Directional derivative representation of $D_g L(g,\xi)$)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$ [Hal00]. Let $L : G \times \mathfrak{g} \to \mathbb{R}$ be smooth and η an arbitrary curve in $\mathfrak{g}$. For all $\delta g \in T_g G$ we regard the partial derivative $D_g L(g,\xi) \in T_g^* G$ as the covector characterized by*

$$\langle D_g L(g,\xi), \delta g\rangle = \langle T_e^* L_g D_g L(g,\xi), \eta\rangle = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(g \exp(\varepsilon\eta), \xi). \tag{5.3}$$

Proof. Since G is a smooth manifold, the differential of the map $L(g,\xi)$ is a linear map $T_gG \to \mathbb{R}$, as ξ is fixed. Therefore the partial derivative $D_gL(g,\xi)$ is an element of the dual space T_g^*G. Moreover, for any $\eta \in \mathfrak{g}$, let $\gamma(\varepsilon) = g\exp(\varepsilon\eta)$ being the smooth curve on the manifold. Then $\gamma(0) = g$ and

$$\begin{aligned}\delta g \;=\; \dot{\gamma}(0) &= \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0}\gamma(\varepsilon) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} g\exp(\varepsilon\eta)\\ &= g\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0}\exp(\varepsilon\eta) = g\,\eta\\ &= T_eL_g(\eta),\end{aligned} \tag{5.4}$$

with δg being the infinitesimal variation of g induced by η for any $\varepsilon \in (-c,c)$ [Lee08]. The corresponding directional derivative is,

$$\langle D_gL(g,\xi),\delta g\rangle = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L(\gamma(\varepsilon),\xi) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L(g\exp(\varepsilon\eta),\xi),$$

using $\langle D_gL(g,\xi),\delta g\rangle = D_gL(g,\xi)(\delta g)$ [Lee08]. Additionally, the directional derivative can be expressed by the pulled-back covector $\mu_g = T_e^*L_gD_gL(g,\xi)$ to

$$\langle D_gL(g,\xi),\delta g\rangle = \langle D_gL(g,\xi),g\eta\rangle = \langle D_gL(g,\xi),T_eL_g(\eta)\rangle = \langle T_e^*L_gD_gL(g,\xi),\eta\rangle,$$

using eq. 5.4. □

5.1.2 Hamiltonian Mechanics

5.1.3 Legendre Transformation

5.1.4 Properties of the Lagrangian Flow

5.2 Unforced 3D-Pendulum

In the following Section, the unforced mechanics of an three degrees of freedom pendulum given by [Lee08] are presented, using Lagrangian mechanics of section 5.1. Following, the momentum preservation properties of the Lagrangian flow of the model will be verified.

5.2.1 Derivation of Lie Group Mechanics

Configuration Manifold The 3D-Pendulum model, illustrated in **??**, consist of a rigid body rotating under the influence of gravity g attached to a fixed frictionless pivot [SSC+04]. Moreover, the system has no external control moment, since it is unforced. A space frame and a body-fixed frame attached to the pendulum, with origin at the pivot are defined. The gravity direction in the space frame is

represented by the unit vector $\boldsymbol{e}_3$, defining the vector from the pivot to the center of mass to $\boldsymbol{\rho_c}$ in the body frame. Following the fixed pivot, there is no translational degree of freedom. Therefore, the configuration manifold of the model is given by the special orthogonal group

$$SO(3) = \{\boldsymbol{R} \in \mathbb{R}^{3\times 3} \| \boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I}, \det \boldsymbol{R} = 1\}.$$

The configuration is completely determined by the rotation matrix $\boldsymbol{R} \in SO(3)$, which maps the body-fixed frame to the space frame. Additionally, in $SO(3)$ the attitude kinematics equation is defined by

$$\dot{\boldsymbol{R}} = \boldsymbol{R}\hat{\boldsymbol{\omega}}, \tag{5.5}$$

with the angular velocity $\boldsymbol{\omega} \in \mathbb{R}^3$, mapped to the Lie algebra $\mathfrak{so}(3)$ by the hat map $\hat{\cdot} : \mathbb{R}^3 \to \mathfrak{so}(3)$ [Lee08]. Consequently, $\hat{\boldsymbol{\omega}}$ is a 3×3 skew symmetric matrix.

Lagrangian To derive the Lagrangian $L : SO(3) \times \mathfrak{so}(3) \to \mathbb{R}$, the kinetic $T : SO(3) \times \mathfrak{so}(3) \to \mathbb{R}$ and potential $U : SO(3) \to \mathbb{R}$ energy of the rigid body are introduced. The rotational inertia of the mass is expressed in the body-frame by the inertia matrix $\boldsymbol{J} \in \mathbb{R}^{3\times 3}$. [Lee08] defines

$$T(\boldsymbol{\omega}) = \int_B \|\boldsymbol{\omega}\hat{\boldsymbol{\rho}}_c\|^2 \, dm(\rho_c) = \frac{1}{2}\operatorname{tr}[\hat{\boldsymbol{\omega}}\boldsymbol{J}_d\hat{\boldsymbol{\omega}}^\top] = \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{J}\boldsymbol{\omega}, \tag{5.6}$$

as a general expression for the kinetic energy for the single body B with the velocity $\boldsymbol{\omega} \times \boldsymbol{\rho}_c = \hat{\boldsymbol{\omega}}\boldsymbol{\rho}_c$, where

$$\boldsymbol{J_d} = \int_B \hat{\boldsymbol{\rho}}_c\hat{\boldsymbol{\rho}}_c^\top \, dm = \frac{1}{2}\operatorname{tr}[\boldsymbol{J}]\,\boldsymbol{I} - \boldsymbol{J}, \tag{5.7}$$

is a nonstandard inertia matrix, with the relation

$$(\boldsymbol{J}\boldsymbol{\omega})^\wedge = \hat{\boldsymbol{\omega}}\boldsymbol{J}_d + \boldsymbol{J}_d\hat{\boldsymbol{\omega}}, \tag{5.8}$$

for any $\boldsymbol{\omega} \in \mathbb{R}^3$. The potential energy is given by

$$U(\boldsymbol{R}) = -mg\,\boldsymbol{e}_3^\top \boldsymbol{R}\boldsymbol{\rho}_c, \tag{5.9}$$

where $\boldsymbol{R}\boldsymbol{\rho}_c$ denotes the vector from the pivot to the center of mass expressed in the space-fixed frame, and $\boldsymbol{e}_3^\top \boldsymbol{R}\boldsymbol{\rho}_c$ is its projection onto the gravity direction $\boldsymbol{e}_3$ [Lee08]. Using 5.6 and 5.9, the Lagrangian of the unforced 3D pendulum is given by

$$L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = T(\hat{\boldsymbol{\omega}}) - U(\boldsymbol{R}) = \frac{1}{2}\operatorname{tr}[\hat{\boldsymbol{\omega}}\boldsymbol{J}_d\hat{\boldsymbol{\omega}}^\top] + mg\,\boldsymbol{e}_3^\top \boldsymbol{R}\boldsymbol{\rho}_c. \tag{5.10}$$

Euler-Lagrange Equations In order to define the Euler-Lagrange Equation 5.1, the derivatives of 5.10 have to be calculated. Firstly, the directional derivative $D_{\hat{\boldsymbol{\omega}}}L(\boldsymbol{R}, \hat{\boldsymbol{\omega}})$ is given by applying 5.2, which yields to

$$\begin{aligned}
\langle D_{\hat{\boldsymbol{\omega}}}L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \delta\hat{\boldsymbol{\omega}}\rangle &= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}} + \epsilon\,\delta\hat{\boldsymbol{\omega}}) \\
&= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} \frac{1}{2}\operatorname{tr}\big[(\hat{\boldsymbol{\omega}} + \epsilon\,\delta\hat{\boldsymbol{\omega}})\,\boldsymbol{J}_d\,(\hat{\boldsymbol{\omega}} + \epsilon\,\delta\hat{\boldsymbol{\omega}})^T\big] \\
&= \frac{1}{2}\operatorname{tr}\big[\delta\hat{\boldsymbol{\omega}}\,\boldsymbol{J}_d\,\hat{\boldsymbol{\omega}}^T + \hat{\boldsymbol{\omega}}\,\boldsymbol{J}_d\,(\delta\hat{\boldsymbol{\omega}})^T\big]\,.
\end{aligned}$$

By using the anti symmetry identity of a cross product $\hat{\boldsymbol{x}} = -\hat{\boldsymbol{x}}^\top$ for any $\boldsymbol{x} \in \mathbb{R}^3$ and $\operatorname{tr}[\boldsymbol{AB}] = \operatorname{tr}[\boldsymbol{BA}]$ for any $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{3\times 3}$, the directional derivative is rewritten as

$$\begin{aligned}
\langle D_{\hat{\boldsymbol{\omega}}}L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \delta\hat{\boldsymbol{\omega}}\rangle &= \frac{1}{2}\operatorname{tr}[-\delta\hat{\boldsymbol{\omega}}\,\boldsymbol{J}_d\,\hat{\boldsymbol{\omega}} - \hat{\boldsymbol{\omega}}\,\boldsymbol{J}_d\,\delta\hat{\boldsymbol{\omega}}] \\
&= \frac{1}{2}\operatorname{tr}\big[-\big(\boldsymbol{J}_d\hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}}\boldsymbol{J}_d\big)\,\delta\hat{\boldsymbol{\omega}}\big] \\
&= \langle \boldsymbol{J}_d\hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}}\boldsymbol{J}_d,\, \delta\hat{\boldsymbol{\omega}}\rangle \\
&= \langle (\boldsymbol{J}\boldsymbol{\omega})^\wedge,\, \delta\hat{\boldsymbol{\omega}}\rangle\,, \qquad (5.11)
\end{aligned}$$

with the definition of the inner product **??** and the skew-symmetric property of 5.8.

Moreover, the derivative $D_{\boldsymbol{R}}L(\boldsymbol{R}, \hat{\boldsymbol{\omega}})$, which is in this model only dependent of the potential energy $U(\boldsymbol{R})$, is calculated by 5.3 to

$$\begin{aligned}
\langle D_{\mathbf{R}}L(\mathbf{R}, \hat{\boldsymbol{\omega}}),\, \delta\mathbf{R}\rangle &= \langle T_e^* L_{\mathbf{R}}\, D_{\mathbf{R}}L(\mathbf{R}, \hat{\boldsymbol{\omega}}),\, \hat{\boldsymbol{\eta}}\rangle \\
&= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} L(\mathbf{R}\exp(\epsilon\hat{\boldsymbol{\eta}}), \hat{\boldsymbol{\omega}}) \\
&= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} mg\,\mathbf{e}_3^\top\mathbf{R}\exp(\epsilon\hat{\boldsymbol{\eta}})\boldsymbol{\rho}_c \\
&= mg\,\mathbf{e}_3^\top\mathbf{R}\hat{\boldsymbol{\eta}}\,\boldsymbol{\rho}_c \\
&= mg\,(\mathbf{R}^\top\mathbf{e}_3)^\top\hat{\boldsymbol{\eta}}\,\boldsymbol{\rho}_c \\
&= mg\,(\boldsymbol{\rho}_c \times \mathbf{R}^\top\mathbf{e}_3)\cdot\boldsymbol{\eta},
\end{aligned}$$

since $\boldsymbol{a}^\top\hat{\boldsymbol{x}}\boldsymbol{b} = \boldsymbol{a}\cdot(\boldsymbol{x}\times\boldsymbol{b}) = (\boldsymbol{b}\times\boldsymbol{a})\cdot\boldsymbol{x}$ for any $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{x} \in \mathbb{R}^3$. Applying the definition of inner products **??**, it follows that

$$\langle T_e^* L_{\mathbf{R}}\, D_{\mathbf{R}}L(\mathbf{R}, \hat{\boldsymbol{\omega}}),\, \hat{\boldsymbol{\eta}}\rangle = \big\langle (mg\,(\boldsymbol{\rho}_c \times \mathbf{R}^\top\mathbf{e}_3))^\wedge, \hat{\boldsymbol{\eta}}\big\rangle\,. \qquad (5.12)$$

Since the directional derivative

$$D_{\hat{\boldsymbol{\omega}}}L(\mathbf{R}, \hat{\boldsymbol{\omega}}) \mathrel{\widehat{=}} \mathbf{J}_d\hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}}\mathbf{J}_d,$$

was derived in 5.11, the coadjoint term in the Euler–Lagrange equations can be written in matrix form by using the identity **??**. Therefore,

$$\begin{aligned} ad^*_{\hat{\boldsymbol{\omega}}} D_{\hat{\boldsymbol{\omega}}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}) &= [\mathbf{J}_d \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \mathbf{J}_d, \hat{\boldsymbol{\omega}}] \\ &= (\mathbf{J}_d \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \mathbf{J}_d)\hat{\boldsymbol{\omega}} - \hat{\boldsymbol{\omega}}(\mathbf{J}_d \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \mathbf{J}_d) \\ &= (\boldsymbol{J}\boldsymbol{\omega} \times \boldsymbol{\omega})^\wedge \end{aligned}$$

with the relation $\hat{\boldsymbol{x}}\hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}\hat{\boldsymbol{x}} = (\boldsymbol{x} \times \boldsymbol{y})^\wedge$ for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$ and 5.8. Furthermore by the anti symmetry of the cross product, it follows that

$$ad^*_{\hat{\boldsymbol{\omega}}} D_{\hat{\boldsymbol{\omega}}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}) = (\boldsymbol{J}\boldsymbol{\omega} \times \boldsymbol{\omega})^\wedge = -(\boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega})^\wedge \tag{5.13}$$

Finally, the Euler-Lagrange equations of the unforced 3D-pendulum are expressed in $\mathfrak{so}(3)$ $SO(3)$ form, by substituting 5.11, 5.12 and 5.13 into 5.1, which result in

$$(\mathbf{J}\dot{\boldsymbol{\omega}})^\wedge + (\boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}))^\wedge = (mg\, \boldsymbol{\rho}_c \times \mathbf{R}^T \mathbf{e}_3)^\wedge, \tag{5.14}$$

$$\dot{\mathbf{R}} = \mathbf{R}\hat{\boldsymbol{\omega}}. \tag{5.15}$$

Legendre Transformation In order to obtain the Hamilton equations for the unforced 3D-pendulum model, the Legendre transformation $\mathbb{F}L : (SO(3) \times \mathfrak{so}(3)) \to (SO(3) \times \mathfrak{so}(3)^*)$ **??** has to be applied, which results in the expression for the momentum

$$\hat{\boldsymbol{\Pi}} = D_{\hat{\boldsymbol{\omega}}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = (\boldsymbol{J}\boldsymbol{\omega})^\wedge \tag{5.16}$$

where $\hat{\boldsymbol{\Pi}} \in \mathfrak{so}(3)^*$. By substituting 5.16 into 5.14 and 5.15, the Hamilton equations of an unforced 3D-pendulum are

$$\dot{\hat{\boldsymbol{\Pi}}} + (\mathbf{J}^{-1}\boldsymbol{\Pi} \times \boldsymbol{\Pi})^\wedge = (mg\, \boldsymbol{\rho}_c \times \mathbf{R}^T \mathbf{e}_3)^\wedge, \tag{5.17}$$

$$\dot{\mathbf{R}} = \mathbf{R}(\mathbf{J}^{-1}\boldsymbol{\Pi})^\wedge. \tag{5.18}$$

Finally, the derived Euler-Lagrange equations 5.14, 5.15 and the Hamilton equations 5.17, 5.18 correspond to the given 3D-pendulum results of [Lee08].

5.2.2 Verification Of Momentum Conservation

The verification of momentum conservation is particularly relevant in the context of this thesis, since the use of Lagrangian mechanics on Lie groups preserves the intrinsic geometric structure of the dynamics, which is essential for accurate simulation and for certifiable trajectory optimization methods [TJVG25]. Therefore, in the following, the momentum conservation property of the continuous 3D pendulum is verified based on its symmetry and Noether's theorem.

[Lee08] identifies a symmetry of the unforced 3D-pendulum's Lagrangian, which is infinitesimally invariant under an action of the group $H = SO(2) \simeq S^1$. Hereby,

a rotation about the gravity direction $\boldsymbol{e}_3$ does not change the Lagrangian. Thus, $SO(2)$ and S^1 describe the same symmetry group under the left action of H on G $\Phi : S^1 \times SO(3) \to SO(3)$

$$\Phi(\theta, \boldsymbol{R}) = \exp_{SO(3)}(\theta \hat{\boldsymbol{e}}_3)\boldsymbol{R}, \tag{5.19}$$

where $\exp_{SO(3)}(\theta \hat{\boldsymbol{e}}_3)$ expresses a finite rotation about the gravity direction $\boldsymbol{e}_3$ by the angle θ [Lee08]. By Noether's Theorem **??**, the infinitesimally invariance of the 3D-pendulum's Lagrangian is verified, if

$$dL(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) \cdot \zeta_{SO(3)\times\mathfrak{so}(3)}(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = 0, \tag{5.20}$$

consequently preserving the Lagrangian momentum map along the flow $J_L\big(\mathcal{F}_L^T(\boldsymbol{R}, \hat{\boldsymbol{\omega}})\big) = J_L(\boldsymbol{R}, \hat{\boldsymbol{\omega}})$.

Therefore, the infinitesimal generator of the lifted group action on $SO(3) \times \mathfrak{so}(3)$ is required [Lee08].With the attitude kinematics relation 5.5, the left trivialization $\phi_L(\mathbf{R}, \dot{\boldsymbol{R}}) = \phi_L(\mathbf{R}, \boldsymbol{R}\hat{\boldsymbol{\omega}}) = (\boldsymbol{R}, T_e L_{\boldsymbol{R}^{-1}} \boldsymbol{R}\hat{\boldsymbol{\omega}}) = (\boldsymbol{R}, \hat{\boldsymbol{\omega}})$, and the left trivialization $\phi_L^{-1}(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = (\boldsymbol{R}, T_e L_{\boldsymbol{R}} \hat{\boldsymbol{\omega}}) = (\mathbf{R}, \mathbf{R}\hat{\boldsymbol{\omega}})$ are taken from **??** and **??**, with the differential T_R of 4.2. Following [Lee08], the lifted infinitesimal generator $\zeta_{SO(3)\times\mathfrak{so}(3)} : SO(3) \times \mathfrak{so}(3) \to T(SO(3) \times \mathfrak{so}(3))$ is computed to

$$\begin{aligned}
\zeta_{SO(3)\times\mathfrak{so}(3)}(\mathbf{R}, \hat{\boldsymbol{\omega}}) &= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \phi_L \circ T_{\mathbf{R}}\Phi_{\exp_{SO(2)}(\epsilon\zeta)} \circ \phi_L^{-1}(\mathbf{R}, \hat{\boldsymbol{\omega}}) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \phi_L \circ T_{\mathbf{R}}\Phi_{\exp_{SO(2)}(\epsilon\zeta)}(\mathbf{R}, \mathbf{R}\hat{\boldsymbol{\omega}}) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \phi_L(\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\mathbf{R}, \exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\mathbf{R}\hat{\boldsymbol{\omega}}) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \big(\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\mathbf{R}, \mathbf{R}^{-1}\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)^{-1}\big(\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\mathbf{R}\hat{\boldsymbol{\omega}}\big)\big) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} (\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\mathbf{R}, \hat{\boldsymbol{\omega}}) \\
&= (\zeta\hat{\boldsymbol{e}}_3\mathbf{R}, \mathbf{0})\,.
\end{aligned} \tag{5.21}$$

This result can be substituted into 5.20, using 5.12, which gives

$$\begin{aligned}
dL(\mathbf{R}, \hat{\boldsymbol{\omega}}) \cdot \zeta_{SO(3)\times\mathfrak{so}(3)}(\mathbf{R}, \hat{\boldsymbol{\omega}}) &= dL(\mathbf{R}, \hat{\boldsymbol{\omega}}) \cdot (\zeta\hat{\boldsymbol{e}}_3\mathbf{R}, \mathbf{0}) \\
&= \langle D_{\mathbf{R}}L(\mathbf{R}, \hat{\boldsymbol{\omega}}),\, \zeta\hat{\boldsymbol{e}}_3\mathbf{R}\rangle + \langle D_{\hat{\boldsymbol{\omega}}}L(\mathbf{R}, \hat{\boldsymbol{\omega}}),\, \mathbf{0}\rangle \\
&= \langle T_e^* L_{\mathbf{R}} D_{\mathbf{R}}L(\mathbf{R}, \hat{\boldsymbol{\omega}}),\, \hat{\boldsymbol{\eta}}\rangle\,.
\end{aligned} \tag{5.22}$$

Since the the lifted generator has no component in the $\hat{\boldsymbol{\omega}}$ direction, the component directional derivative $D_{\hat{\boldsymbol{\omega}}}$ equals zero in zero direction. Moreover, by the definition of the lifted infinitesimal generator **??**, the result of 5.21 redefines the variational relationship being

$$(\delta\boldsymbol{R}, \delta\hat{\boldsymbol{\omega}}) = (\zeta\hat{\boldsymbol{e}}_3\mathbf{R}, \mathbf{0}). \tag{5.23}$$

In consequence, with the new relation of $\delta\boldsymbol{R} = \zeta\hat{\boldsymbol{e}}_3\boldsymbol{R} = \boldsymbol{R}\hat{\boldsymbol{\eta}}$, the variation $\hat{\boldsymbol{\eta}}$ can be expressed as

$$\hat{\boldsymbol{\eta}} = \zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{R}. \tag{5.24}$$

This can be substituted into 5.22, following

$$\left\langle T_e^* L_{\mathbf{R}} D_{\mathbf{R}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}),\, \zeta\mathbf{R}^{-1}\hat{\boldsymbol{e}}_3\mathbf{R}\right\rangle = \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} L\big(\mathbf{R}\exp\big(\epsilon\zeta\mathbf{R}^{-1}\hat{\boldsymbol{e}}_3\mathbf{R}\big)\,, \hat{\boldsymbol{\omega}}\big)\,.$$

Since only the potential term of the 3D-pendulum $U(\boldsymbol{R})$ 5.9 depends on $\mathbf{R}$ and 5.24 is skew symmetric, it can be shown that

$$\begin{aligned}
\left.\frac{d}{d\epsilon}\right|_{\epsilon=0} L\big(\mathbf{R}\exp\big(\epsilon\zeta\mathbf{R}^{-1}\hat{\boldsymbol{e}}_3\mathbf{R}\big)\,, \hat{\boldsymbol{\omega}}\big) &= mg\,\boldsymbol{e}_3^T\mathbf{R}\left(\zeta\mathbf{R}^{-1}\hat{\boldsymbol{e}}_3\mathbf{R}\right)\boldsymbol{\rho}_c \\
&= mg\,(\boldsymbol{\rho}_c^\top\left(\zeta\mathbf{R}^{-1}\hat{\boldsymbol{e}}_3\mathbf{R}\right)^\top\mathbf{R}^\top\boldsymbol{e}_3)^\top \\
&= -\,mg\,\boldsymbol{\rho}_c^\top\left(\zeta\mathbf{R}^{-1}\hat{\boldsymbol{e}}_3\mathbf{R}\right)\mathbf{R}^\top\boldsymbol{e}_3 \\
&= -\,mg\,\boldsymbol{\rho}_c^\top\zeta\mathbf{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{e}_3 \\
&= -\,mg\,\boldsymbol{\rho}_c^\top\zeta\mathbf{R}^{-1}(\boldsymbol{e}_3\times\boldsymbol{e}_3) \\
&= 0,
\end{aligned} \tag{5.25}$$

because $\boldsymbol{e}_3\times\boldsymbol{e}_3 = \mathbf{0}$ and $\boldsymbol{R}^{-1} = \boldsymbol{R}^\top$ for any $\boldsymbol{R}\in SO(3)$. Therefore, the condition of 5.20 is verified and the Lagrangian is infinitesimally invariant under the symmetry action. Consequently, by Noether's Theorem **??**, the corresponding momentum map $J_L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = \boldsymbol{e}_3^T\boldsymbol{R}\mathbf{J}\hat{\boldsymbol{\omega}}$ of the continuous 3D-pendulum, which is given in [Lee08], is preserved along the Lagrangian flow

$$J_L\big(\mathcal{F}_L^T(\mathbf{R}, \hat{\boldsymbol{\omega}})\big) = J_L(\mathbf{R}, \hat{\boldsymbol{\omega}}).$$

Chapter 6

Discrete Mechanics For Rigid Bodies On Lie Groups

6.1 Discrete Lagrangian Mechanics

6.1.1 Discrete Euler-Lagrange Equations

Proposition 4 (Directional derivative representation of $D_{g_k}L_d(g_k, f_k)$)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$ [Hal00]. Let $L_d : G \times G \to \mathbb{R}$ be smooth and η_k an arbitrary element of $\mathfrak{g}$. For all $\delta g_k \in T_{g_k}G$ with $\delta g_k = T_e L_{g_k}(\eta_k)$* **??***, we regard the partial derivative $D_{g_k}L_d(g_k, f_k) \in T^*_{g_k}G$ as the covector characterized by*

$$\langle D_{g_k}L_d(g_k, f_k), \delta g_k\rangle = \langle T^*_e L_{g_k} D_{g_k}L_d(g_k, f_k), \eta_k\rangle = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(g_k \exp(\varepsilon\eta_k), f_k). \quad (6.1)$$

Proof. Since G is a smooth manifold, the differential of the map $L_d(g_k, f_k)$ is a linear map $T_{g_k}G \to \mathbb{R}$, as f_k is fixed. Therefore, the partial derivative $D_{g_k}L_d(g_k, f_k)$ is an element of the dual space $T^*_{g_k}G$. Moreover, for any $\eta_k \in \mathfrak{g}$, let

$$\gamma(\varepsilon) = g_k \exp(\varepsilon\eta_k)$$

be a smooth curve on the manifold. Then $\gamma(0) = g_k$ and

$$\begin{aligned} \delta g_k \;=\; \dot\gamma(0) &= \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \gamma(\varepsilon) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} g_k \exp(\varepsilon\eta_k) \\ &= g_k \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \exp(\varepsilon\eta_k) = g_k\,\eta_k \\ &= T_e L_{g_k}(\eta_k), \end{aligned} \quad (6.2)$$

with δg_k being the infinitesimal variation of g_k induced by η_k for any $\varepsilon \in (-c, c)$ [Lee08]. The corresponding directional derivative is

$$\langle D_{g_k}L_d(g_k, f_k), \delta g_k\rangle = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(\gamma(\varepsilon), f_k) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(g_k \exp(\varepsilon\eta_k), f_k),$$

using $\langle D_{g_k} L_d(g_k, f_k), \delta g_k \rangle = D_{g_k} L_d(g_k, f_k)(\delta g_k)$ [Lee08]. Additionally, the directional derivative can be expressed by the pulled-back covector

$$\mu_k = T_e^* L_{g_k} D_{g_k} L_d(g_k, f_k)$$

to obtain

$$\begin{aligned}\langle D_{g_k} L_d(g_k, f_k), \delta g_k \rangle &= \langle D_{g_k} L_d(g_k, f_k), g_k \eta_k \rangle \\ &= \langle D_{g_k} L_d(g_k, f_k), T_e L_{g_k}(\eta_k) \rangle \\ &= \langle T_e^* L_{g_k} D_{g_k} L_d(g_k, f_k), \eta_k \rangle,\end{aligned}$$

using eq. 6.2. □

Proposition 5 (Directional derivative representation of $D_{f_k} L_d(g_k, f_k)$)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$ [Hal00]. Let $L_d : G \times G \to \mathbb{R}$ be smooth and X_k an arbitrary element of $\mathfrak{g}$. For all $\delta f_k \in T_{f_k} G$ with $\delta f_k = T_e L_{f_k}(X_k)$* **??***, we regard the partial derivative $D_{f_k} L_d(g_k, f_k) \in T_{f_k}^* G$ as the covector characterized by*

$$\langle D_{f_k} L_d(g_k, f_k), \delta f_k \rangle = \langle T_e^* L_{f_k} D_{f_k} L_d(g_k, f_k), X_k \rangle = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(g_k, f_k \exp(\varepsilon X_k)). \tag{6.3}$$

Proof. Since G is a smooth manifold, the differential of the map $L_d(g_k, f_k)$ is a linear map $T_{f_k} G \to \mathbb{R}$, as g_k is fixed. Therefore, the partial derivative $D_{f_k} L_d(g_k, f_k)$ is an element of the dual space $T_{f_k}^* G$. Moreover, for any $X_k \in \mathfrak{g}$, let

$$\gamma(\varepsilon) = f_k \exp(\varepsilon X_k)$$

be a smooth curve on the manifold. Then $\gamma(0) = f_k$ and

$$\begin{aligned}\delta f_k \;=\; \dot{\gamma}(0) &= \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \gamma(\varepsilon) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} f_k \exp(\varepsilon X_k) \\ &= f_k \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \exp(\varepsilon X_k) = f_k\, X_k \\ &= T_e L_{f_k}(X_k),\end{aligned} \tag{6.4}$$

with δf_k being the infinitesimal variation of f_k induced by X_k for any $\varepsilon \in (-c, c)$ [Lee08]. The corresponding directional derivative is,

$$\langle D_{f_k} L_d(g_k, f_k), \delta f_k \rangle = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(g_k, \gamma(\varepsilon)) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(g_k, f_k \exp(\varepsilon X_k)),$$

using $\langle D_{f_k} L_d(g_k, f_k), \delta f_k \rangle = D_{f_k} L_d(g_k, f_k)(\delta f_k)$ [Lee08]. Additionally, the directional derivative can be expressed by the pulled-back covector

$$\mu_k = T_e^* L_{f_k} D_{f_k} L_d(g_k, f_k)$$

to obtain

$$\begin{aligned}\langle D_{f_k} L_d(g_k, f_k), \delta f_k\rangle &= \langle D_{f_k} L_d(g_k, f_k), f_k X_k\rangle \\ &= \langle D_{f_k} L_d(g_k, f_k), T_e L_{f_k}(X_k)\rangle \\ &= \langle T_e^* L_{f_k} D_{f_k} L_d(g_k, f_k), X_k\rangle,\end{aligned}$$

using eq. 6.4. □

6.1.2 Forced Discrete Euler-Lagrange Equations

6.1.3 Discrete Hamiltonian Mechanics

6.1.4 Discrete Legendre Transformation

6.1.5 Properties of the Discrete Lagrangian Flow

6.1.6 Computational Implementation

In contrast to the continuous mechanics of **??**, numerical simulations can be performed for discrete-time mechanical model representations. Therefore, a simulation framework has to be defined. To determine the relative attitude update $\boldsymbol{F}_k \in SO(3)$, the implicit Euler-Lagrange **??** and Hamilton equations **??** have to be solved at every time step k. Therefore, an efficient computational algorithm is required to for practical numerical simulations, especially for complex multibody-mechanical models. [Lee08] proposes an approach of expressing the group element $\boldsymbol{F}_k \in G$ on the linear vector space of the Lie algebra, by using the exponential map. Thereby, the implicit equation is solved numerically by a Newton iteration [MW01] in $\mathbb{R}^3$.

Firstly, the implicit discrete-time Euler-Lagrange equations or the discrete-time Hamilton equations have to be converted into an equivalent vector equation in $\mathbb{R}^3$, for example, using the Rodrigures formula

$$\boldsymbol{F} = \exp(\hat{\boldsymbol{f}}) = \boldsymbol{I}_{3\times 3} + \frac{\sin\|\boldsymbol{f}\|}{\|\boldsymbol{f}\|}\,\hat{\boldsymbol{f}} + \frac{1-\cos\|\boldsymbol{f}\|}{\|\boldsymbol{f}\|^2}\,\hat{\boldsymbol{f}}^2, \tag{6.5}$$

for $f \in \mathbb{R}^3$ [MR99]. However, it requires the evaluation of expensive sin and cos terms, which increase the computational time and costs. In order to imrpove the computational efficiency, the exponential map 6.5 is replaced by the Cayley transformation [Ise01].

Cayley Transformation The Cayley transformation is a local diffeomorphism between $SO(3) \to \mathbb{R}^3$, which is defined as

$$F = \mathrm{Cay}(f) = (\boldsymbol{I} - \hat{\boldsymbol{f}})(\boldsymbol{I} + \hat{\boldsymbol{f}})^{-1} = \frac{1}{1+\boldsymbol{f}^\top \boldsymbol{f}}((1-\boldsymbol{f}^\top \boldsymbol{f})\boldsymbol{I} + 2\hat{\boldsymbol{f}} + 2\boldsymbol{f}\boldsymbol{f}^\top, \tag{6.6}$$

representing an rotation θ given by $\|f\| = \tan\left(\frac{\theta}{2}\right)$ [BLdD23]. Therefore, the Cayley transformation consists of the constraint $\theta < \pi$, which requires the step size h to be chosen sufficiently small. After substituting the Cayley transformation 6.6 into the implicit equations **??** or **??**, a equivalent vector equation is obtained

$$0 = A(\boldsymbol{f}), \tag{6.7}$$

where $A : \mathbb{R}^3 \to \mathbb{R}^3$ [CJ12].

Newton Iteration For solving the implicit vector equation 6.7 Newton's method is used [MW01]. The iteration rule is given by

$$\boldsymbol{f}^{(i+1)} = \boldsymbol{f}^{(i)} - \nabla A\left(\boldsymbol{f}^{(i)}\right)^{-1} A\left(\boldsymbol{f}^{(i)}\right), \tag{6.8}$$

where convergence to the solution is reached until $\left\|f^{(i+1)} - f^{(i)}\right\| < \epsilon$ for a defined ϵ. A initial guess $\boldsymbol{f}^0$ can be obtained by linearizing the vector equation 6.7 [Lee08]. Finally, the rotation matrix $\boldsymbol{F}_k \in SO(3)$ at k is calculated using the Cayley transformation 6.6.

6.2 Unforced Discrete Mechanics

6.2.1 3D-Pendulum

In the following Section, the unforced mechanics of an three degrees of freedom pendulum given by [Lee08] are presented, using discrete Lagrangian mechanics of section 6.1. This will correspond to the equivalent discrete case of the continuous unforced 3D-pendulum of section 5.2. Following, numerical simulations are performed to verify the momentum conservation and constant energy of the derived mechanical systems equations.

Configuration Manifold The configuration manifold of the unforced 3D-pendulum in a discrete case, consits equal settings of the pendulums space and body-fixed frame. The gravity direction in the space frame is represented by the unit vector $\boldsymbol{e}_3$, defining the vector from the pivot to the center of mass to $\boldsymbol{\rho_c} = \frac{l}{2}\boldsymbol{e}_3^b$ in the body frame. As in the continuous formulation of Section 5.2, the discrete configuration manifold is $SO(3)$. Therefore, the attitude of the 3D-pendulum is described by a rotation matrix $\boldsymbol{R}_k \in SO(3)$, which maps the body-fixed frame to the reference frame at the discrete time step k. [Lee08] defines the discrete-time attitude kinematics equation as

$$\boldsymbol{R}_{k+1} = \boldsymbol{R}_k \boldsymbol{F}_k, \tag{6.9}$$

with the rotation matrix $\boldsymbol{F}_k$, representing the relative attitude update between two integration steps. Additionally, [Lee08] introduces adjoint operator relations for

$\boldsymbol{F} \in SO(3)$ and the skew-symmetric matrix $\hat{\boldsymbol{\eta}} \in \mathfrak{so}(3)$, which are given as

$$\mathrm{Ad}_{\boldsymbol{F}}\, \hat{\boldsymbol{\eta}} = \boldsymbol{F}\, \hat{\boldsymbol{\eta}}\, \boldsymbol{F}^\top = (\boldsymbol{F}\boldsymbol{\eta})^\wedge, \tag{6.10}$$

$$\mathrm{Ad}^*_{\boldsymbol{F}}\, \hat{\boldsymbol{\eta}} = \boldsymbol{F}^\top\, \hat{\boldsymbol{\eta}}\, \boldsymbol{F} = (\boldsymbol{F}^\top\boldsymbol{\eta})^\wedge. \tag{6.11}$$

Discrete Lagrangian As introduced in **??**, the discrete Lagrangian $L_d(\boldsymbol{R}_k, \boldsymbol{F}_k) : SO(3) \times SO(3) \to \mathbb{R}$ is an approximation of the action integral of the continuous Lagrangian over one time step h. In order to discretize the given continuous Lagrangian of 5.10

$$L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = T(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) - U(\boldsymbol{R}) = \frac{1}{2}\,\mathrm{tr}[\hat{\boldsymbol{\omega}}\boldsymbol{J}_d\hat{\boldsymbol{\omega}}^\top] + mg\,\boldsymbol{e}_3^\top\boldsymbol{R}\boldsymbol{\rho}_c, \tag{6.12}$$

the time dependent angular velocity $\hat{\boldsymbol{\omega}}(t)$ and rotation matrix $\boldsymbol{R}(t)$ have to be represented on the discrete time interval $[t_k, t_{k+1}]$ [MW01]. Given the definition of the exponential for an $n \times n$ matrix $\boldsymbol{X}$

$$\exp(\boldsymbol{X}) = \sum_{m=0}^{\infty} \frac{\boldsymbol{X}^m}{m},$$

[Hal00], its first-order expansion can be written as

$$\exp(\boldsymbol{X}) = \boldsymbol{I} + \boldsymbol{X} + O(\boldsymbol{X}^2). \tag{6.13}$$

Therefore, for small h, the exponential map $\boldsymbol{F}_k = \exp(h\hat{\boldsymbol{\omega}}_k)$ can be approximated using 6.13 to

$$\boldsymbol{F}_k = \exp(h\hat{\boldsymbol{\omega}}_k) \approx \boldsymbol{I} + h\hat{\boldsymbol{\omega}}_k,$$

following the general discrete approximation of the angular velocity

$$\hat{\boldsymbol{\omega}}_k \approx \frac{1}{h}(\boldsymbol{F}_k - \boldsymbol{I}), \tag{6.14}$$

[Lee08]. Secondly, quadrature rules, for example, the midpoint rule and the trapezoidal rule can be applied to create an integrator $F^{L_d} : (\boldsymbol{R}_0, \boldsymbol{F}_0) \to (\boldsymbol{R}_1, \boldsymbol{F}_1)$, or the symplectic integration method of Strömer-Verlet [MW01]. In order to align with [Lee08], the trapezodial rule is applied for the approximation for an integral, given by

$$\int_{t_k}^{t_{k+1}} f(t)\,dt \approx \frac{h}{2}\left(f(t_k) + f(t_{k+1})\right), \tag{6.15}$$

for $h = t_{k+1} - t_k$ [But16]. Applying 6.15 to the Lagrangian 6.12, it follows

$$\begin{aligned}\int_{t_k}^{t_{k+1}} U(\boldsymbol{R})\,dt = \int_{t_k}^{t_{k+1}} mg\,\boldsymbol{e}_3^\top\boldsymbol{R}\boldsymbol{\rho}_c\,dt &\approx \frac{h}{2}\,mg\,\boldsymbol{e}_3^\top\boldsymbol{R}_k\boldsymbol{\rho}_c + \frac{h}{2}\,mg\,\boldsymbol{e}_3^\top\boldsymbol{R}_{k+1}\boldsymbol{\rho}_c \\ &= \frac{h}{2}\,mg\,\boldsymbol{e}_3^\top\boldsymbol{R}_k\boldsymbol{\rho}_c + \frac{h}{2}\,mg\,\boldsymbol{e}_3^\top\boldsymbol{R}_k\boldsymbol{F}_k\boldsymbol{\rho}_c, \\ &= U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k),\end{aligned} \tag{6.16}$$

using 6.9, as only the potential energy $U(\boldsymbol{R})$ of the continuous Lagrangian depends on $\boldsymbol{R}$. Substituting 6.14 and 6.16 into 6.12, the discrete Lagrangian is calculated to

$$\begin{aligned} L_d(\boldsymbol{R}_k, \boldsymbol{F}_k) &= \frac{1}{2h}\operatorname{tr}\big[(\boldsymbol{F}_k - \boldsymbol{I})\boldsymbol{J}_d(\boldsymbol{F}_k - \boldsymbol{I})^\top\big] - \frac{h}{2}\left(U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k)\right) \\ &= \frac{1}{2h}\operatorname{tr}\big[\boldsymbol{F}_k\boldsymbol{J}_d\boldsymbol{F}_k^\top - \boldsymbol{J}_d\boldsymbol{F}_k^\top - \boldsymbol{F}_k\boldsymbol{J}_d + \boldsymbol{J}_d\big] - \frac{h}{2}\left(U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k)\right) \\ &= \frac{1}{2h}\operatorname{tr}[2\boldsymbol{J}_d - 2\boldsymbol{F}_k\boldsymbol{J}_d] - \frac{h}{2}\left(U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k)\right) \\ &= \frac{1}{h}\operatorname{tr}[(\boldsymbol{I} - \boldsymbol{F}_k)\boldsymbol{J}_d] - \frac{h}{2}\left(U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k)\right), \end{aligned} \tag{6.17}$$

using $\operatorname{tr}[\boldsymbol{F}_k\boldsymbol{J}_d\boldsymbol{F}_k^\top] = \operatorname{tr}[\boldsymbol{J}_d\boldsymbol{F}_k^\top\boldsymbol{F}_k] = \operatorname{tr}[\boldsymbol{J}_d]$ and $\operatorname{tr}[\boldsymbol{J}_d\boldsymbol{F}_k^\top] = \operatorname{tr}[\boldsymbol{F}_k\boldsymbol{J}_d]$, since $\boldsymbol{J}_d$ is defined as an diagonal matrix $\boldsymbol{J}_d = \boldsymbol{J}_d^\top$ [Lee08].

Discrete Euler-Lagrange Equations In order to define the Euler-Lagrange Equation **??**, the derivatives of 6.12 have to be calculated. Firstly, the directional derivative $\langle D_{\boldsymbol{F}_k}L(\boldsymbol{R}_k, \boldsymbol{F}_k), \delta\boldsymbol{f}_k\rangle = \left\langle T_e^*L_{\boldsymbol{F}_k}\, D_{\boldsymbol{F}_k}L_d(\boldsymbol{R}_k, \boldsymbol{F}_k), \hat{\boldsymbol{X}}_k\right\rangle$ is given by applying 6.3, which yields to

$$\begin{aligned} \left\langle T_e^*L_{\boldsymbol{F}_k}\, D_{\boldsymbol{F}_k}L_d(\boldsymbol{R}_k, \boldsymbol{F}_k), \hat{\boldsymbol{X}}_k\right\rangle &= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} L_d\Big(\boldsymbol{R}_k, \boldsymbol{F}_k\exp(\epsilon\hat{\boldsymbol{X}}_k)\Big) \\ &= -\frac{1}{h}\operatorname{tr}\Big[\boldsymbol{F}_k\hat{\boldsymbol{X}}_k\boldsymbol{J}_d\Big] + \frac{h}{2}mg\,\boldsymbol{e}_3^\top\boldsymbol{R}_k\boldsymbol{F}_k\hat{\boldsymbol{X}}_k\boldsymbol{\rho}_c \\ &= -\frac{1}{h}\operatorname{tr}\Big[\hat{\boldsymbol{X}}_k\boldsymbol{J}_d\boldsymbol{F}_k\Big] + \frac{h}{2}mg\,\Big((\boldsymbol{R}_k\boldsymbol{F}_k)^\top\boldsymbol{e}_3\Big)^\top\hat{\boldsymbol{X}}_k\boldsymbol{\rho}_c, \end{aligned}$$

using the trace identity $\operatorname{tr}[\boldsymbol{ABC}] = \operatorname{tr}[\boldsymbol{BCA}]$ for any $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{R}^{n\times n}$. Additionally, [Lee08] introduces the relation $\operatorname{tr}[\hat{\boldsymbol{s}}\boldsymbol{B}] = -\left\langle \boldsymbol{B} - \boldsymbol{B}^\top, \hat{\boldsymbol{s}}\right\rangle$ for any $\boldsymbol{s} \in \mathbb{R}^3, \boldsymbol{B} \in \mathbb{R}^{3\times 3}$. Applying it together with $\boldsymbol{a}^\top\hat{\boldsymbol{x}}\boldsymbol{b} = (\boldsymbol{b}\times\boldsymbol{a})\cdot\boldsymbol{x}$ for any $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{x} \in \mathbb{R}^3$, the inner product defeinition **??**, and the linearity of the pairing $\langle\boldsymbol{A}, \boldsymbol{B}\rangle + \langle\boldsymbol{C}, \boldsymbol{B}\rangle = \langle\boldsymbol{A} + \boldsymbol{C}, \boldsymbol{B}\rangle$ for any $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{R}^{n\times n}$, the final directional derivative of the discrete Lagrangian respect to $\boldsymbol{F}_k$ is calculated to

$$\begin{aligned} \left\langle T_e^*L_{\boldsymbol{F}_k}\, D_{\boldsymbol{F}_k}L_d(\boldsymbol{R}_k, \boldsymbol{F}_k), \hat{\boldsymbol{X}}_k\right\rangle &= \left\langle \frac{1}{h}\Big(\boldsymbol{J}_d\boldsymbol{F}_k - \boldsymbol{F}_k^\top\boldsymbol{J}_d\Big), \hat{\boldsymbol{X}}_k\right\rangle + \left\langle \frac{h}{2}mg\,\Big(\boldsymbol{\rho}_c\times(\boldsymbol{R}_k\boldsymbol{F}_k)^\top\boldsymbol{e}_3\Big)^\wedge, \hat{\boldsymbol{X}}_k\right\rangle \\ &= \left\langle \frac{1}{h}\Big(\boldsymbol{J}_d\boldsymbol{F}_k - \boldsymbol{F}_k^\top\boldsymbol{J}_d\Big) + \Big(\frac{h}{2}mg\,\boldsymbol{\rho}_c\times\boldsymbol{R}_{k+1}^\top\boldsymbol{e}_3\Big)^\wedge, \hat{\boldsymbol{X}}_k\right\rangle. \end{aligned}$$

Therefore,

$$T_e^*L_{\boldsymbol{F}_k}\, D_{\boldsymbol{F}_k}L_d(\boldsymbol{R}_k, \boldsymbol{F}_k) \mathrel{\widehat{=}} \frac{1}{h}\Big(\boldsymbol{J}_d\boldsymbol{F}_k - \boldsymbol{F}_k^\top\boldsymbol{J}_d\Big) + \frac{h}{2}\hat{\boldsymbol{M}}_{k+1}, \tag{6.18}$$

with the definition of

$$\boldsymbol{M}_k = mg\,\boldsymbol{\rho}_c\times\boldsymbol{R}_k^\top\boldsymbol{e}_3, \tag{6.19}$$

being the moment due to the potential energy in body frame [Lee08].

Moreover, the directional derivative

$$\left\langle D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}), \delta \boldsymbol{R}_{k+1} \right\rangle = \left\langle T_e^* L_{\boldsymbol{R}_{k+1}} D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle,$$

which is in this model only dependent of the potential energy $U(\boldsymbol{R}_{k+1})+U(\boldsymbol{R}_{k+1}\boldsymbol{F}_{k+1})$, is calculated by 6.1 to

$$\begin{aligned}
\left\langle T_e^* L_{\boldsymbol{R}_{k+1}} D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle &= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} L_d(\boldsymbol{R}_{k+1} \exp(\epsilon \hat{\boldsymbol{\eta}}_{k+1}), \boldsymbol{F}_{k+1}) \\
&= \frac{h}{2} mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_{k+1} \hat{\boldsymbol{\eta}}_{k+1} \boldsymbol{\rho}_c + \frac{h}{2} mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_{k+1} \hat{\boldsymbol{\eta}}_{k+1} \boldsymbol{F}_{k+1} \boldsymbol{\rho}_c \\
&= \frac{h}{2} mg \left[\left(\boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3 \right) + \left(\boldsymbol{F}_{k+1} \boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3 \right) \right] \cdot \boldsymbol{\eta}_{k+1}
\end{aligned}$$

Applying the definition of inner products **??**, it follows

$$\begin{aligned}
\left\langle T_e^* L_{\boldsymbol{R}_{k+1}} D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle &= \left\langle \frac{h}{2} mg \left[\left(\boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3 \right) + \left(\boldsymbol{F}_{k+1} \boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3 \right) \right]^\wedge, \hat{\boldsymbol{\eta}}_{k+1} \right\rangle. \\
&= \left\langle \frac{h}{2} \left(\hat{\boldsymbol{M}}_{k+1} + (\boldsymbol{F}_{k+1} \boldsymbol{M}_{k+2})^\wedge \right), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle,
\end{aligned}$$

which implies that the derivative of 6.12 with repsect to $\boldsymbol{R}_{k+1}$ is given to

$$D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}) = \frac{h}{2} \left[\hat{\boldsymbol{M}}_{k+1} + (\boldsymbol{F}_{k+1} \boldsymbol{M}_{k+2})^\wedge \right]. \tag{6.20}$$

Since the directional derivative $T_e^* L_{\boldsymbol{F}_k} D_{\boldsymbol{F}_k} L_d(\boldsymbol{R}_k, \boldsymbol{F}_k)$ was derived in 6.18, the coadjoint term in the discrete Euler-Lagrange equation **??** can be written in matrix form to

$$\begin{aligned}
\mathrm{Ad}^*_{\boldsymbol{F}_{k+1}^{-1}} \left(T_e^* L_{\boldsymbol{F}_{k+1}} D_{\boldsymbol{F}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}) \right) &= \boldsymbol{F}_{k+1} \left[\frac{1}{h} \left(\boldsymbol{J}_d \boldsymbol{F}_{k+1} - \boldsymbol{F}_{k+1}^\top \boldsymbol{J}_d \right) + \frac{h}{2} \hat{\boldsymbol{M}}_{k+2} \right] \boldsymbol{F}_{k+1}^\top \\
&= \frac{1}{h} \left(\boldsymbol{F}_{k+1} \boldsymbol{J}_d - \boldsymbol{J}_d \boldsymbol{F}_{k+1}^\top \right) + \frac{h}{2} \boldsymbol{F}_{k+1} \hat{\boldsymbol{M}}_{k+2} \boldsymbol{F}_{k+1}^\top \\
&= \frac{1}{h} \left(\boldsymbol{F}_{k+1} \boldsymbol{J}_d - \boldsymbol{J}_d \boldsymbol{F}_{k+1}^\top \right) + \frac{h}{2} (\boldsymbol{F}_{k+1} \boldsymbol{M}_{k+2})^\wedge,
\end{aligned} \tag{6.21}$$

by using the identity 6.11.

Finally, the Euler-Lagrange equations of the unforced 3D-pendulum are expressed in $\mathfrak{so}(3)$ $SO(3)$ form, substituting 6.18, 6.20 and 6.21 into **??**, which results in

$$\frac{1}{h} \left(\boldsymbol{F}_{k+1} \boldsymbol{J}_d - \boldsymbol{J}_d \boldsymbol{F}_{k+1}^\top - \boldsymbol{J}_d \boldsymbol{F}_k + \boldsymbol{F}_k^\top \boldsymbol{J}_d \right) = h\, \hat{\boldsymbol{M}}_{k+1}, \tag{6.22}$$

$$\boldsymbol{R}_{k+1} = \boldsymbol{R}_k \boldsymbol{F}_k. \tag{6.23}$$

[Lee08] describes the procedure of obtaining the discrete-time flow map $(\boldsymbol{R}_k, \boldsymbol{R}_{k+1}) \to (\boldsymbol{R}_{k+1}, \boldsymbol{R}_{k+2})$, by solving 6.22 finding $\boldsymbol{F}_{k+1}$ for given $(\boldsymbol{R}_k, \boldsymbol{R}_{k+1})$ and $\boldsymbol{F}_k = \boldsymbol{R}_k^\top \boldsymbol{R}_{k+1}$. Secondly, $\boldsymbol{R}_{k+2}$ is derived from 6.23.

Discrete Legendre Transformation In order to obtain the discrete Hamilton equations for the unforced 3D-pendulum model, the discrete Legendre transformation $\mathbb{F}^+L_d, \mathbb{F}^-L_d : (SO(3) \times SO(3)) \to (SO(3) \times \mathfrak{so}(3)^*)$ with $\mathbb{F}^+L_d(\boldsymbol{R}_{k-1}, \boldsymbol{F}_{k-1}) = \mathbb{F}^-L_d(\boldsymbol{R}_k, \boldsymbol{F}_k)$ **??** have to be applied, which results in the expression for the discrete momentum

$$\begin{aligned}
\hat{\boldsymbol{\Pi}}_k &= -\frac{h}{2}\left(\hat{\boldsymbol{M}}_k + (\boldsymbol{F}_k\boldsymbol{M}_{k+1})^\wedge\right) + \frac{1}{h}\left(\boldsymbol{F}_k\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_k^\top\right) + \frac{h}{2}\left(\boldsymbol{F}_k\boldsymbol{M}_{k+1}\right)^\wedge \\
&= \frac{1}{h}\left(\boldsymbol{F}_k\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_k^\top\right) - \frac{h}{2}\hat{\boldsymbol{M}}_k, && (6.24)\\
\hat{\boldsymbol{\Pi}}_{k+1} &= \frac{1}{h}\left(\boldsymbol{J}_d\boldsymbol{F}_k - \boldsymbol{F}_k^\top\boldsymbol{J}_d\right) + \frac{h}{2}\hat{\boldsymbol{M}}_{k+1}. && (6.25)
\end{aligned}$$

where $\hat{\boldsymbol{\Pi}} \in \mathfrak{so}(3)^*$. Either, by substituting 6.24 and 6.25 into 6.22 and 6.23, or by applying **??**, the discrete Hamilton equations of an unforced 3D-pendulum are calculated to

$$\begin{aligned}
\frac{1}{h}\left(\boldsymbol{F}_k\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_k^\top\right) &= \hat{\boldsymbol{\Pi}}_k + \frac{h}{2}\hat{\boldsymbol{M}}_k, && (6.26)\\
\boldsymbol{R}_{k+1} &= \boldsymbol{R}_k\boldsymbol{F}_k, && (6.27)\\
\hat{\boldsymbol{\Pi}}_{k+1} &= (\boldsymbol{F}_k^\top\boldsymbol{\Pi}_k)^\wedge + \frac{h}{2}(\boldsymbol{F}_k^\top\boldsymbol{M}_k)^\wedge + \frac{h}{2}\hat{\boldsymbol{M}}_{k+1}. && (6.28)
\end{aligned}$$

Finally, the discrete map $(\boldsymbol{R}_k, \boldsymbol{\Pi}_k) \to (\boldsymbol{R}_{k+1}, \boldsymbol{\Pi}_{k+1})$ is obtained by solving implicit the equation 6.26 for $\boldsymbol{F}_k$ with given $(\boldsymbol{R}_k, \boldsymbol{\Pi}_k)$. In consequence, $\boldsymbol{R}_{k+1}$ is derived from 6.26, and $\boldsymbol{\Pi}_{k+1}$ from 6.28.

6.3 Forced Discrete Mechanics

6.3.1 Acrobot

Chapter 7

Evaluation

7.1 Numerical Simulation Results

In this section, numerical simulations are conducted to evaluate the correctness of the derived discrete integrators for mechanical models on Lie groups. In particular, the structure-preserving properties of the variational discretizations derived in Section **??** are examined. The analysis focuses especially on the preservation of the momentum map associated with the symmetry of the system, the evolution on the Lie-group configuration manifold, and the long-time behaviour of the total energy. This validation is important since discrete mechanical models obtained by variational integration provide a reliable basis for certifiable trajectory optimization. If the discrete flow violates the manifold structure or introduces artificial drift in conserved quantities such as momentum, the resulting optimization model becomes less faithful to the underlying mechanics [Lee08]. In contrast, variational integrators are constructed from a discrete variational principle and are symplectic, momentum preserving, and consist robust long-time energy behaviour [MW01]. For the evaluation, the evolution of the total energy, the momentum-map behaviour, and the preservation of the Lie-group structure of the unforced 3D-pendulum are compared with the classical fourth-order Runge-Kutta method and a projected Runge-Kutta on $SO(3)$ as reference integrators [CO03, Hai00] (see Table. 7.1).

Table 7.1: Comparison of the considered integration methods properties.

Method	**Symplectic**	**Manifold Preserving**
Variational Integrator	✓	✓
RK4	✗	✗
Projected RK4	✗	✓

Runge–Kutta 4th order As a basline integrator, the classical explicit fourth-order Runge-Kutta (RK4) is applied to the continuous rigid-body equations written

as as first-order system $(\dot{\boldsymbol{R}}, \dot{\boldsymbol{\omega}}) = \boldsymbol{f}(\boldsymbol{R}, \boldsymbol{\omega})$. For a system of the form

$$\dot{\boldsymbol{y}} = \boldsymbol{f}(\boldsymbol{y}), \tag{7.1}$$

the classical RK4 scheme computes the four stage values

$$(\boldsymbol{k}_1^R, \boldsymbol{k}_1^\omega) = \boldsymbol{f}(\boldsymbol{R}_n, \boldsymbol{\omega}_n), \tag{7.2}$$

$$(\boldsymbol{k}_2^R, \boldsymbol{k}_2^\omega) = \boldsymbol{f}\left(\boldsymbol{R}_n + \frac{h}{2}\boldsymbol{k}_1^R, \boldsymbol{\omega}_n + \frac{h}{2}\boldsymbol{k}_1^\omega\right), \tag{7.3}$$

$$(\boldsymbol{k}_3^R, \boldsymbol{k}_3^\omega) = \boldsymbol{f}\left(\boldsymbol{R}_n + \frac{h}{2}\boldsymbol{k}_2^R, \boldsymbol{\omega}_n + \frac{h}{2}\boldsymbol{k}_2^\omega\right), \tag{7.4}$$

$$(\boldsymbol{k}_4^R, \boldsymbol{k}_4^\omega) = \boldsymbol{f}\left(\boldsymbol{R}_n + h\boldsymbol{k}_3^R, \boldsymbol{\omega}_n + h\boldsymbol{k}_3^\omega\right), \tag{7.5}$$

and updates the numerical solution according to

$$\boldsymbol{R}_{n+1} = \boldsymbol{R}_n + \frac{h}{6}\left(\boldsymbol{k}_1^R + 2\boldsymbol{k}_2^R + 2\boldsymbol{k}_3^R + \boldsymbol{k}_4^R\right), \tag{7.6}$$

$$\boldsymbol{\omega}_{n+1} = \boldsymbol{\omega}_n + \frac{h}{6}\left(\boldsymbol{k}_1^\omega + 2\boldsymbol{k}_2^\omega + 2\boldsymbol{k}_3^\omega + \boldsymbol{k}_4^\omega\right), \tag{7.7}$$

for each discrete-time step n [But16].

The explicit RK4 method does not preserve the orthogonality constraint $\boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I}$ and leaves the manifold configuration of $SO(3)$, since the rotation matrix $\boldsymbol{R}_n$ is updated additively in the vector space of $\mathbb{R}^{3\times 3}$ [CO03].

RK4 projected to $SO(3)$ To satisfy the orthogonality constraint

$$SO(3) = \left\{\boldsymbol{R} \in \mathbb{R}^{3\times 3} \;\middle|\; \boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I},\ \det(\boldsymbol{R}) = 1\right\}.$$

the explicit RK4 method is combined with a projection of the discrete-time updated rotation matrix $\boldsymbol{R}_n$ on $SO(3)$. Therefore, after performing the RK4 step in $\mathbb{R}^{3\times 3}$ the obtained rotation matrix $\widetilde{\boldsymbol{R}}_{n+1}$, the singular value decomposition is used

$$\widetilde{\boldsymbol{R}}_{n+1} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top, \tag{7.8}$$

following a reconstruction to derive the updated rotation matrix

$$\boldsymbol{R}_{n+1} = \boldsymbol{U}\boldsymbol{V}^\top, \tag{7.9}$$

which is projected on $SO(3)$ [YTT11]. Consequently, $\boldsymbol{R}_{n+1}$ satisfy the orthogonality constraint, while the angular velocity $\boldsymbol{\omega}_{n+1}$ remains unchanged on the vector space $\mathbb{R}^{3\times 3}$. However, this method does not inherit structure preserving properties of the Lie group variational integrator, for example, symplecticity or momentum conservation [CO03].

7.1.1 Unforced 3D Pendulum

For evaluating the geometric properties preservation of the previously defined unforced 3D-pendulum (see Section **??**), numerical simulations are conducted. The rigid body is modeled as a uniform solid cylinder of length $l = 1\,\mathrm{m}$ and radius $r = 5\,\mathrm{cm}$, with its center of mass at $\frac{l}{2}$. Thereby, the physical parameters characterizing the configuration manifold, are chosen as

$$m = 1\,\mathrm{kg}, \qquad g = 9.81\,\mathrm{m\,s^{-2}}, \tag{7.10}$$

$$\boldsymbol{\rho}_c = \begin{bmatrix} 0 & 0 & 0.5 \end{bmatrix}^\top \mathrm{m}, \qquad \boldsymbol{e}_3 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^\top. \tag{7.11}$$

The inertia matrix about the center of mass can be taken from the inertia table given in [Ric]. Here, the rotation of the center of mass in x and y direction are represented about the central diameter $\boldsymbol{J}_{11,com} = \boldsymbol{J}_{22,com} = \frac{1}{12}m(3r^2 + 4l^2)$, and the rotation of the center of mass in gravity direction z is perpendicular to the axis of rotation $\boldsymbol{J}_{33,com} = \frac{1}{2}mr^2$. The off-diagonal terms are equal to zero, because the body-fixed frame is aligned with the principal axes of the cylindrically symmetric body [Lee08]. Additionally, the inertia matrix about the pivot in body-frame is obtained by the parallel axis theorem of [BR18] given by

$$\mathbf{J} = \mathbf{J}_{\mathrm{com}} + m\left(\|\boldsymbol{\rho}_c\|^2\mathbf{I} - \boldsymbol{\rho}_c\boldsymbol{\rho}_c^\top\right). \tag{7.12}$$

After substituting the chosen values, the 3D-pendulum's inertia matrix is given as

$$\mathbf{J} = \begin{bmatrix} 0.33396 & 0 & 0 \\ 0 & 0.33396 & 0 \\ 0 & 0 & 0.00125 \end{bmatrix} \mathrm{kg\,m^2}. \tag{7.13}$$

Given 5.7, the non-standard inertia matrix is calculated to

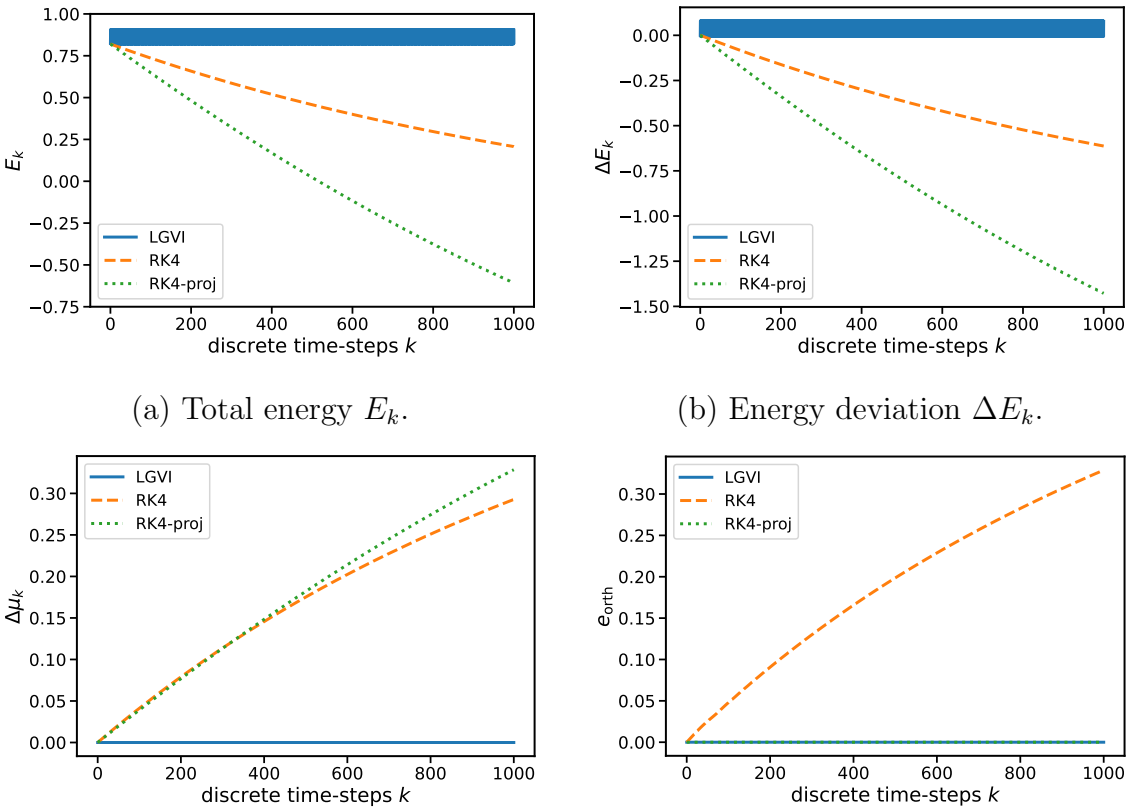
$$\mathbf{J}_d = \begin{bmatrix} 0.000625 & 0 & 0 \\ 0 & 0.000625 & 0 \\ 0 & 0 & 0.333333 \end{bmatrix} \mathrm{kg\,m^2}. \tag{7.14}$$

Using the introduced computational approach of Section 6.1.6, the derived discrete-time implicit Hamilton equation 6.26 of the 3D-pendulum is converted to an equivalent vector equation in $\mathbb{R}^3$ using the Cayley transformation. From [CJ12], the corresponding nonlinear vector equation $0 = \boldsymbol{A}(\boldsymbol{f})$ of the 3D-pendulum is taken as

$$0 = \boldsymbol{a} + \boldsymbol{a} \times \boldsymbol{f} + \boldsymbol{f}(\boldsymbol{a}^\top\boldsymbol{f}) - 2\boldsymbol{J}\boldsymbol{f} = \boldsymbol{A}(f), \tag{7.15}$$

with $\boldsymbol{a} = h(\boldsymbol{\Pi}_k + \frac{h}{2}\boldsymbol{M}_k)$. Additionally, [CJ12] defines its Jacobian $\nabla A(\boldsymbol{f})$ to

$$\nabla A(\boldsymbol{f}) = \hat{\boldsymbol{a}} + (\boldsymbol{a}^\top\boldsymbol{f})\mathbf{I} + \boldsymbol{f}\boldsymbol{a}^\top - 2\mathbf{J}. \tag{7.16}$$



(a) Total energy E_k.

(b) Energy deviation ΔE_k.

(c) Momentum error $\Delta\mu_k$.

(d) Orthogonality error e_{orth}.

Figure 7.1: Comparison of the numerical preservation properties for the unforced 3D pendulum.

The tolerance for the Newton iteration is defined to $\left\| f^{(i+1)} - f^{(i)} \right\| < 10^{-12}$. The initial conditions are taken from [Lee08]

$$\boldsymbol{f}^0 = (2\boldsymbol{J} - \hat{\boldsymbol{a}})^{-1}\boldsymbol{a}, \qquad \mathbf{R}_0 = \mathbf{I}, \qquad \boldsymbol{\Omega}_0 = \begin{bmatrix} 4.14 & 4.14 & 4.14 \end{bmatrix}^\top \mathrm{rad\,s^{-1}}, \tag{7.17}$$

where $\boldsymbol{f}^0$ is the linearized solution of 7.15. The simulation is conducted over $[0, 1000]$ discrete-time steps and $h = 0.05\,\mathrm{s}$. The corresponding implementation is documented in the GitHub repository [Els26].

Fig. 7.1 illustrates the results of the numerical simulation of the different methods of Table 7.1, by plotting the corresponding total energy

$$E_k = \frac{1}{2}\boldsymbol{\Omega}_k^\top \mathbf{J}\boldsymbol{\Omega}_k - mg\,\boldsymbol{e}_3^\top \mathbf{R}_k \boldsymbol{\rho}_c,$$

and its deviation $\Delta E_k = E_k - E_0$, the angular momentum error along the gravity direction

$$\Delta\mu_k = \boldsymbol{e}_3^\top \mathbf{R}_k \boldsymbol{\Pi}_k - \boldsymbol{e}_3^\top \mathbf{R}_0 \boldsymbol{\Pi}_0,$$

defined in [Lee08], and the orthogonality error $e_{\mathrm{orth},k} = \left\|\mathbf{I} - \mathbf{R}_k^\top \mathbf{R}_k\right\|_F$.

The total energy (see Fig. 7.1 (a)) and the corresponding energy deviation (see Fig. 7.1 (b)) illustrates the preserving energy behavior over a long-time in discrete-time flow for the Lie Group Variational Integrator (LGVI). It remains close to the reference level with only small bounded oscillations, because of their symplectic strcuture, which are the exact solution of a pertubed Hamiltonian [Hai94]. However, the RK4 and projected RK4 display a clear drift of the energy away from its real value, which remains constant for this model, as the system is unforced.

Moreover, the momentum error (see Fig. 7.1 (c)) confirms that the LGVI method preserves the momentum, validated for continuous time in 5.2.2. The computational results of RK4 and its projection on $SO(3)$ show that these methods are not reliable drifting from the momentum initial value, as they not preserve the total linear momentum [Lee08]. Especially, the projected RK4 consits of a stronger discrepancy, as it fixes $\boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I}$, while the angular velocity ω remains unchanged, which influences the momentum.

Lastly, the orthogonality error highlights that the structure of the Lie group configuration manifold is well preserved in the LGVI, which is depictured in Fig. 7.1 (d). Similar result is obtained from the projected RK4, while the classic explicit RK4 violates the orthogonality constraint. For RK4 e_{orth} increases linearly with respect to the simulation time-steps.

Consequently, only the LGVI preserves the manifold structure together with the momentum and energy behaviour over a long-time horizon. While RK4 can be projected to the Lie group manifold of $SO(3)$, it can not preserve the momentum and consits of drifting energy behaviour over long-time.

7.2 Optimization

$$\min_{\substack{\{R_k\}_{k=0}^{N} \\ \{F_k\}_{k=0}^{N-1}, \{u_{p,k}\}_{k=1}^{N-1}}} \quad \rho_f \|R_N - R_N^{\text{des}}\|_F^2 + \sum_{k=0}^{N-1} \left(\alpha_R \|R_k - R_N^{\text{des}}\|_F^2\right) + \sum_{k=1}^{N-1} \gamma^{-1} \|u_{p,k}\|_2^2, \tag{7.18a}$$

$$\text{s.t.} \quad \delta_2 f_{32,k+1} - \delta_3 f_{23,k+1} - \delta_3 f_{32,k} + \delta_2 f_{23,k} - h^2(M_{1,k+1} + u_{1,k+1}) = 0, \tag{7.18b}$$

$$\delta_3 f_{13,k+1} - \delta_1 f_{31,k+1} - \delta_1 f_{13,k} + \delta_3 f_{31,k} - h^2(M_{2,k+1} + u_{2,k+1}) = 0, \tag{7.18c}$$

$$\delta_1 f_{21,k+1} - \delta_2 f_{12,k+1} - \delta_2 f_{21,k} + \delta_1 f_{12,k} - h^2(M_{3,k+1} + u_{3,k+1}) = 0, \tag{7.18d}$$

$$(R_{k+1})_{ij} - \sum_{\ell=1}^{3} (R_k)_{i\ell} (F_k)_{\ell j} = 0, \tag{7.18e}$$

$$\|r_{1,k}\|_2^2 - 1 = \|r_{2,k}\|_2^2 - 1 = \|r_{3,k}\|_2^2 - 1 = 0, \tag{7.18f}$$

$$r_{1,k}^\top r_{2,k} = r_{2,k}^\top r_{3,k} = r_{1,k}^\top r_{3,k} = 0, \tag{7.18g}$$

$$r_{1,k} \times r_{2,k} - r_{3,k} = r_{2,k} \times r_{3,k} - r_{1,k} = r_{3,k} \times r_{1,k} - r_{2,k} = 0_{3\times 1}, \tag{7.18h}$$

$$\|f_{1,k}\|_2^2 - 1 = \|f_{2,k}\|_2^2 - 1 = \|f_{3,k}\|_2^2 - 1 = 0, \tag{7.18i}$$

$$f_{1,k}^\top f_{2,k} = f_{2,k}^\top f_{3,k} = f_{1,k}^\top f_{3,k} = 0, \tag{7.18j}$$

$$f_{1,k} \times f_{2,k} - f_{3,k} = f_{2,k} \times f_{3,k} - f_{1,k} = f_{3,k} \times f_{1,k} - f_{2,k} = 0_{3\times 1}, \tag{7.18k}$$

$$\frac{\text{tr}(F_k) - 1}{2} \geq f_{c,\min}, \tag{7.18l}$$

$$u_{\max}^2 - \|u_{p,k}\|_2^2 \geq 0. \tag{7.18m}$$

$$\text{with} \qquad M_{k+1} = \begin{bmatrix} M_{1,k+1} \\ M_{2,k+1} \\ M_{3,k+1} \end{bmatrix} = mg \begin{bmatrix} \rho_2 r_{33,k+1} - \rho_3 r_{32,k+1} \\ \rho_3 r_{31,k+1} - \rho_1 r_{33,k+1} \\ \rho_1 r_{32,k+1} - \rho_2 r_{31,k+1} \end{bmatrix}, \tag{7.19}$$

$$u_{k+1} = \begin{bmatrix} u_{1,k+1} \\ u_{2,k+1} \\ u_{3,k+1} \end{bmatrix} = \begin{bmatrix} r_{32,k+1} u_{p3,k+1} - r_{33,k+1} u_{p2,k+1} \\ r_{33,k+1} u_{p1,k+1} - r_{31,k+1} u_{p3,k+1} \\ r_{31,k+1} u_{p2,k+1} - r_{32,k+1} u_{p1,k+1} \end{bmatrix}, \tag{7.20}$$

$$R_k = [r_{1,k}\ r_{2,k}\ r_{3,k}] \in \mathbb{R}^{3\times 3}, \qquad r_{1,k}, r_{2,k}, r_{3,k} \in \mathbb{R}^{3\times 1}, \tag{7.21}$$

$$F_k = [f_{1,k}\ f_{2,k}\ f_{3,k}] \in \mathbb{R}^{3\times 3}, \qquad f_{1,k}, f_{2,k}, f_{3,k} \in \mathbb{R}^{3\times 1}, \tag{7.22}$$

$$J_d = \operatorname{diag}(\delta_1, \delta_2, \delta_3), \qquad \rho_c = \begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{bmatrix}, \qquad u_{p,k} = \begin{bmatrix} u_{p1,k} \\ u_{p2,k} \\ u_{p3,k} \end{bmatrix}. \tag{7.23}$$

In (7.18b)–(7.18d), the discrete Euler–Lagrange constraints hold for $k = 0, \ldots, N-2$. The kinematic constraint (7.18e) holds for $i, j = 1, 2, 3$ and $k = 0, \ldots, N-1$. The constraints (7.18f)–(7.18h) hold for $k = 0, \ldots, N$, while (7.18i)–(7.18k), (7.18l), and (7.18m) hold for $k = 0, \ldots, N-1$.

For each stage $k = 0, \ldots, N-1$, the decision variables are

$$R_k \in \mathbb{R}^{3\times 3}, \qquad F_k \in \mathbb{R}^{3\times 3}, \qquad u_{p,k} \in \mathbb{R}^3.$$

Hence, each stage contributes

$$9 + 9 + 3 = 21$$

scalar decision variables. Since the terminal attitude $R_N \in \mathbb{R}^{3\times 3}$ is also optimized, the total number of scalar decision variables is

$$9(N+1) + 9N + 3N = 21N + 9.$$

If the fixed initial values $R_0 = \bar{R}_0$ and $F_0 = \bar{F}_0$ are eliminated from the decision vector in the implementation, the number of free scalar decision variables reduces to

$$21N + 9 - 18 = 21N - 9.$$

From (7.18a)

$$\begin{aligned} & p_f \sum_{i=1}^{3} \sum_{j=1}^{3} \big(r_{ij,N} - r_{ij,N}^{\mathrm{des}}\big)^2 \\ & \quad + \sum_{k=0}^{N-1} \alpha_R \sum_{i=1}^{3} \sum_{j=1}^{3} \big(r_{ij,k} - r_{ij,N}^{\mathrm{des}}\big)^2 \\ & \quad + \sum_{k=1}^{N-1} \gamma^{-1} \big(u_{p1,k}^2 + u_{p2,k}^2 + u_{p3,k}^2\big). \end{aligned} \tag{7.24}$$

From (7.18e)

$$r_{11,k+1} - r_{11,k}f_{11,k} - r_{12,k}f_{21,k} - r_{13,k}f_{31,k} = 0, \tag{7.25}$$
$$r_{12,k+1} - r_{11,k}f_{12,k} - r_{12,k}f_{22,k} - r_{13,k}f_{32,k} = 0, \tag{7.26}$$
$$r_{13,k+1} - r_{11,k}f_{13,k} - r_{12,k}f_{23,k} - r_{13,k}f_{33,k} = 0, \tag{7.27}$$
$$r_{21,k+1} - r_{21,k}f_{11,k} - r_{22,k}f_{21,k} - r_{23,k}f_{31,k} = 0, \tag{7.28}$$
$$r_{22,k+1} - r_{21,k}f_{12,k} - r_{22,k}f_{22,k} - r_{23,k}f_{32,k} = 0, \tag{7.29}$$
$$r_{23,k+1} - r_{21,k}f_{13,k} - r_{22,k}f_{23,k} - r_{23,k}f_{33,k} = 0, \tag{7.30}$$
$$r_{31,k+1} - r_{31,k}f_{11,k} - r_{32,k}f_{21,k} - r_{33,k}f_{31,k} = 0, \tag{7.31}$$
$$r_{32,k+1} - r_{31,k}f_{12,k} - r_{32,k}f_{22,k} - r_{33,k}f_{32,k} = 0, \tag{7.32}$$
$$r_{33,k+1} - r_{31,k}f_{13,k} - r_{32,k}f_{23,k} - r_{33,k}f_{33,k} = 0. \tag{7.33}$$

From (7.18h)

$$r_{21,k}r_{32,k} - r_{31,k}r_{22,k} - r_{13,k} = 0, \tag{7.34}$$
$$r_{31,k}r_{12,k} - r_{11,k}r_{32,k} - r_{23,k} = 0, \tag{7.35}$$
$$r_{11,k}r_{22,k} - r_{21,k}r_{12,k} - r_{33,k} = 0, \tag{7.36}$$
$$r_{22,k}r_{33,k} - r_{32,k}r_{23,k} - r_{11,k} = 0, \tag{7.37}$$
$$r_{32,k}r_{13,k} - r_{12,k}r_{33,k} - r_{21,k} = 0, \tag{7.38}$$
$$r_{12,k}r_{23,k} - r_{22,k}r_{13,k} - r_{31,k} = 0, \tag{7.39}$$
$$r_{23,k}r_{31,k} - r_{33,k}r_{21,k} - r_{12,k} = 0, \tag{7.40}$$
$$r_{33,k}r_{11,k} - r_{13,k}r_{31,k} - r_{22,k} = 0, \tag{7.41}$$
$$r_{13,k}r_{21,k} - r_{23,k}r_{11,k} - r_{32,k} = 0. \tag{7.42}$$

From (7.18.k)

$$f_{21,k}f_{32,k} - f_{31,k}f_{22,k} - f_{13,k} = 0, \tag{7.43}$$
$$f_{31,k}f_{12,k} - f_{11,k}f_{32,k} - f_{23,k} = 0, \tag{7.44}$$
$$f_{11,k}f_{22,k} - f_{21,k}f_{12,k} - f_{33,k} = 0, \tag{7.45}$$
$$f_{22,k}f_{33,k} - f_{32,k}f_{23,k} - f_{11,k} = 0, \tag{7.46}$$
$$f_{32,k}f_{13,k} - f_{12,k}f_{33,k} - f_{21,k} = 0, \tag{7.47}$$
$$f_{12,k}f_{23,k} - f_{22,k}f_{13,k} - f_{31,k} = 0, \tag{7.48}$$
$$f_{23,k}f_{31,k} - f_{33,k}f_{21,k} - f_{12,k} = 0, \tag{7.49}$$
$$f_{33,k}f_{11,k} - f_{13,k}f_{31,k} - f_{22,k} = 0, \tag{7.50}$$
$$f_{13,k}f_{21,k} - f_{23,k}f_{11,k} - f_{32,k} = 0. \tag{7.51}$$

7.3 Experimental Results

A thesis almost always has an experimental part, typically some comparison to other approaches. Benchmarking takes time, for running the experiments, but also for thinking them up in the first place, and for analysing the results. Plan accordingly to spend enough time here!

Think about what makes sense to measure, what you want to learn from your measurements. Think about what is really the relevant contribution of your thesis, and how you can prove that you have achieved your goals. Think about what you can measure in order to get a good insight into the performance of various aspects of your design, how you can distinguish between systematic and accidental effects, how you can convince yourself that your results are right. If you get surprising results, do not just say "surprise, surprise, performance isn't as good as hoped". Find out why. Surprises without explanation indicate either that you are clueless about what is going on, or that you have made a mistake. Unconvincing results, therefore, tend to imply unconvincing marks.

Statistics: Measurements always have statistical (sampling) errors. Owing to the deterministic nature of simulations these are sometimes very small, as disturbing factors can be designed. However, the reader should be given an indication of how statistically significant the results are. This is done by providing at least a standard deviation in addition to averages. Whenever the results of several runs are averaged, a standard deviation can (and must) be supplied. After all, you average to reduce statistical errors.

The reproducibility argument applies here just as much as for the implementation. Give enough detail on what you measure, and how you measure it, so that someone who has your implementation (but not your test code) or has re-done your implementation independently, should be able to repeat your measurements and arrive at essentially the same results. In some cases, results seem outright wrong in a thesis. In those cases, not enough detail is provided to allow the supervisor/reader to pinpoint the likely source of the error. Often the cause is systematic errors resulting from an incorrect measurement technique. If it seems wrong, and the text does not convince the reader that it is not wrong, the reader will assume that it is wrong.

Chapter 8

Discussion

Discuss and explain your results. Show how they support your thesis (or, if they do not, give a convincing explanation). It is important to separate objective facts clearly from their discussion (which is bound to contain subjective opinion). If the reader does not understand your results, reconsider if you have managed to extract the core information and explain it in a straightforward way.

Chapter 9

Conclusion

Do not just leave it at the discussion: discuss what you/the reader can learn from the results. Draw some real conclusions. Separate discussion/interpretation of the results clearly from the conclusions you draw from them. (So-called "conclusion creep" tends to upset reviewers. It means surrendering your scientific objectivity.) Identify all shortcomings/limitations of your work, and discuss how they could be fixed ("future work"). It is not a sign of weakness of your work if you clearly analyze and state the limitations. Informed readers will notice them anyway and draw their own conclusions if not addressed properly.

Recap: do not stick to this structure at all cost. Also, remember that the thesis must be:

- honest, stating clearly all limitations
- self–contained, do not write just for the locals, do not assume that the reader has read the same literature as you, do not let the reader work out the details for themselves

This chapter is followed by the list of figures and the bibliography. If you are using acronyms, listing them (with the expanded full name) before the bibliography is also a good idea. The acronyms package helps with consistency and an automatic listing.

Appendix A

BUSTED! Last chance to actually read the HowTo-Section!

Do not forget to check our thesis-guidelines before submitting your *final* thesis.

A.1 Spellcheck reminder

Before handing in your thesis, even for an intermediate review, please perform a proper spellcheck and correct grammar mistakes. Otherwise, there is a high likelihood that your supervisor will return your report immediately without making any comments.

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Acronyms

BIBO	bounded-input bounded-output
HRC	human-robot collaboration
MPC	model-predictive control

Notation

$\mathbf{x}_n$	n-dimensional vector named x
$\mathbf{X}_{m\times n}$	$m \times n$ dimensional matrix named X

List of Symbols

$\mathbf{u}$	control input vector
$\mathbf{u}_k$	control input vector with time step
$\mathbf{x}_k$	state vector with time step

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